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Deep learning-enabled cherry price forecasting and real-time system deployment across multi-market supply chains in India

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For high-value perishable goods such as cherries, accurate price forecasting is crucial for the supply chain, stabilizing producer income, informing policy interventions & real-time decision support at various stakeholder levels. This study enquires whether Deep Learning (DL) architectures can be implemented in real time for daily advisory tools while simultaneously surpassing conventional statistical and Machine Learning (ML) models in predicting cherry prices. To tackle this issue, daily cherry price data from five wholesale markets in India were utilized for the period 2012–2024 to evaluate six forecasting methodologies, i.e., Seasonal Auto-regressive Integrated Moving Average (SARIMA), Prophet, Random Forest (RF), Extreme Gradient Boosting (XGBoost), Long Short-Term Memory (LSTM), and Transformer. The results show that deep learning models are better at capturing non-linear price changes than statistical or tree-based methods. The LSTM and Transformer models consistently performed better across all error metrics (MAE, RMSE, sMAPE, MFE, NMBE, and DA). During the 2025 cherry season, the best-performing LSTM was used as a live, web-based forecasting system that accepted real-time submissions from market officials and produced daily predictions based on field conditions. After the season, we checked the actual prices against the forecasted values and found that they were correct with more than 92 per cent accuracy for market-variety-grade combinations. In major markets like Azadpur, Narwal, and Parimpora, the sMAPE was often below 5–10 per cent, and the error margins (MAE 5–8; RMSE 8–12) were low. Further, the Diebold–Mariano (DM) test showed that deep learning was statistically superior to the baselines. The current study presents a novel and comprehensive operational methodology for advanced Artificial Intelligence (AI) models in the agriculture & allied sectors of India, establishing a scalable framework for incorporating ML/DL into agricultural market intelligence and advisory systems.

Cherry (*Prunus avium*) is a highly valued horticultural crop cultivated globally for its flavor and nutritional value. The top three producers in the world are Iran, the United States of America, and Turkey, with India ranked 26th[1]. The majority of India's cherry production comes from the Himalayan state of Jammu and Kashmir (J&K)[2]. Cherry production in J&K has grown from about 3.8 thousand tonnes to nearly 12 thousand tonnes over the past three decades, reaching a peak of 13 thousand tonnes in 2013[2] and further increasing to around 15.5 thousand tonnes in 2019–2020[3]. Nearly 9% of the Gross State Domestic Product (GSDP) comes from the horticulture sector, and premium Kashmiri cherries are a significant cash crop and exportable commodity[2]. The global and regional importance of cherry cultivation emphasizes the need for efficient market mechanisms and price stability for farmers and traders alike.

However, cherries are seasonal and perishable; weather-related supply shocks or market gluts often cause price volatility. The commodity is extremely sensitive to supply and demand dynamics in real time, and these fluctuations are made worse by the absence of extended storage options[4]. Such market volatilities present serious obstacles for policymakers seeking stable agricultural markets and have a direct impact on farmer incomes and consumer affordability[5]. As a result, timely and accurate price forecasting tools are essential and of the need of the hour. Forecasting helps optimize inventory and logistics, plan for harvest & marketing, and provides the government with real-time information for successful policy implementation[5,6].

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Motivation

In the past, researchers have extensively used time-series modeling techniques to address price volatility in perishables. Because of their ease of use and strong statistical properties, techniques such as the Auto-Regressive Integrated Moving Average (ARIMA) and its seasonal variant (SARIMA) have been used for a long time to forecast agricultural prices. These models have also been widely used across crop types, from perishables such as tomatoes & mangoes to staples such as rice & maize. They are good at capturing trend and seasonality in univariate datasets[7–10]. However, when applied to highly non-linear or abruptly fluctuating data, the predictive performance of these models deteriorates[11,12]. While some improvements are provided by variations such as ARIMA + GARCH, they still rely on linear assumptions that may not fully capture the complexity of commodity price behavior. Although hybrid designs that use machine-learning techniques, such as random forests, for lag selection and combine ARIMA and LSTM models have increased robustness for volatile agricultural price series, they still depend on fixed parametric structures and are susceptible to regime changes[13]. Likewise, data science pipelines that combine traditional time-series models with modern forecasting tools for agricultural products indicate that we need to move away from univariate specifications towards more complex, feature-aware systems[14].

Recent research has moved towards the use of artificial intelligence (AI) and machine learning (ML) techniques that do not require prespecified functional forms to address these constraints. More flexibility in identifying nonlinear patterns in agricultural time series has been demonstrated by Artificial Neural Networks (ANNs), Support Vector Machines (SVMs), Random Forests (RFs), and Gradient Boosting frameworks such as XGBoost[9–12]. Weng et al.[12] showed, for example, that a neural network trained on a sizable dataset of horticultural prices performed better than a tuned ARIMA model over several forecasting horizons. Similarly, XGBoost has been effectively used to forecast rice production, greatly surpassing conventional methods[15,16]. Building on these developments, deep learning models have become popular for forecasting in agriculture, especially Recurrent Neural Networks (RNNs) and LSTM architectures[17,18]. For instance, recent research shows that deep learning architectures like LSTM and GRU can perform much better than traditional benchmarks for agricultural commodity prices, especially when nonlinearity and moderate volatility are strong[19]. These models have demonstrated strong performance in price and yield forecasting and are particularly good at learning long-term dependencies from large, high-dimensional datasets[20–22]. However, they have drawbacks like high processing costs, a need for a lot of data, and difficulties in capturing long-range dependencies or handling sparse data[23].

By addressing some of the drawbacks of RNNs, transformer-based models that use attention mechanisms have opened new possibilities for time-series forecasting[24,25]. Transformers are especially helpful for capturing irregular seasonal trends and multi-year cycles in agricultural prices because they can model longer temporal sequences without depending on recurrence. Recent Meta-transformer frameworks that use Transformer encoders and metaheuristic optimization for hyperparameter tuning show even greater gains for perishable or seasonal agricultural goods. This indicates that attention-based architectures could work well in this area[4]. There is still a noticeable research gap in applying ML & DL techniques to temperate fruit markets, despite the expanding corpus of literature on agricultural price forecasting using these techniques. High-value & seasonal fruits like cherries have been overlooked in previous research, especially in the regional horticulture sectors like J&K, which have mainly concentrated on staple cereals or commonly traded vegetables. Targeted forecasting models that account for the specific market dynamics, perishability, and regional concentration of cherry cultivation in J&K are desperately needed. Mixture-of-experts architectures for forecasting vegetable prices, which dynamically choose specialized predictors across different types of goods, also show that ensemble and attention-inspired designs can deal with cross-commodity heterogeneity better than any one model[26].

Therefore, the design, benchmarking, and operationalization of a real-time price forecasting framework for cherry markets is the specific goal of this study. Also, it focuses on answering the research question of whether DL architectures can offer better accuracy, robustness, and grade-wise fidelity than statistical and ML baselines. Moreover, the study aims to predict that DL models will not only perform better in prediction but also demonstrate operational dependability when applied to real-world market settings. Specifically, the research creates a decision-support system that directly connects AI-driven market intelligence to physical agricultural practices and policies by integrating real-time validation and providing farmers and other stakeholders with daily price forecasts as value-added advisory services. Previous AI-enabled price-forecasting studies in agriculture have relied mainly on ex post evaluation (based on historical data) or produced coarse, low-frequency outputs. To bridge this gap between methodological benchmarking and actionable decision support, the current robust & novel ML/DL based methodological architecture was developed and tested, which helps in (i) providing market-variety-grade-specific forecasts at daily frequency, (ii) validating model superiority with both conventional error metrics and DM test, and (iii) operationalising the best model in a live deployment that ingests official market entries and publishes farmer-facing advisories in real time.

Our work

By creating and evaluating an extensive real-time price forecasting pipeline for cherry markets in Jammu & Kashmir, this study addresses several crucial gaps. The novelty of our system has contributed to the following:

1. *Comparative, grade-aware modeling:* Using a unified evaluation protocol (MAE, RMSE, sMAPE, DA, MFE, NMBE) and DM tests for statistically rigorous model comparisons, we systematically compare six models, viz., SARIMA, Prophet, RF, XGBoost, LSTM, and Transformer across five wholesale markets and three commercial grades.
2. *First real-time, farmer-facing deployment in India:* The best model is built into a full-stack pipeline (FastAPI + PostgreSQL + CI/CD) that takes in daily entries from market officials and posts market–variety–grade–

specific forecasts on the MIC website (www.micskuast.in) while also allowing dissemination through manual reports, social media platforms, and news bulletins. This lets farmers, traders, and planners use the forecasts every day.

3. *Live-season validation and operational robustness:* The system generates and stores daily forecasts during the cherry season, which are then verified against actual market prices. In field settings, the estimates showed low error magnitudes and high directional accuracy, i.e., practical reliability beyond offline backtests.

Through this multi-tier structure, the system establishes a standard for agricultural market intelligence for the perishable fruit industry by combining retrospective modeling with live operational deployment. In addition to improving forecast accuracy, the developed system provides a scalable, replicable approach to enhance price stability and well-informed decision-making in agricultural markets

Related studies

Traditional models for perishable fruit price forecasting

Early methods for predicting the prices of perishable fruits (and other agricultural commodities) relied primarily on econometric and traditional time-series models. Because of their adaptability in simulating trends and seasonality, ARIMA models, in particular, have become the standard. Fruit price series have been frequently analyzed using ARIMA and its variants, such as seasonal ARIMA and ARIMAX with exogenous inputs, which often yied accurate short-term forecasts[27,28]. For instance, Niu et al.[29] used an ARIMA model for short-term price prediction of Hebei pear and found it effective for capturing the seasonal patterns of fruit prices. In the Indian context, Pardhi et al.[9] demonstrated that an ARIMA model could reasonably forecast mango prices during the peak season in a significant market, achieving a one-step-ahead forecast with Mean Absolute Percentage Error (MAPE) below 10%.

These results confirmed that ARIMA can capture the seasonal spike (e.g., price doubling at the season start) with acceptable accuracy. Other statistical models, such as regression with seasonal dummies and volatility models (e.g., ARCH/GARCH), have also been explored to handle price fluctuations. For example, a study[28] on apple prices in Himachal Pradesh compared ARIMA with a GARCH extension to account for heteroskedasticity, and found that ARIMA(6,1,1) with GARCH(1,1) provided the best fit to the historical price series. Although these conventional models are easy to understand, they often fall short of capturing the intricate, nonlinear dynamics of markets for perishable produce because they assume linear or predefined structures already exist in the datasets. Classical models only partially address the weather, disease, and supply gluts that cause sudden price swings in seasonal fruits and vegetables (e.g., via seasonal terms or differencing). In recent years, this constraint has enabled more sophisticated predictive techniques.

It is important to remember that traditional models are still competitive in some situations. An ARIMA model, for example, performed better than an LSTM network for predicting the prices of certain vegetables (such as lotus root and greens), particularly in capturing abrupt swings and linear trends, according to a recent comparative study[30]. The authors noted that the simpler ARIMA model produced fewer errors in those datasets, despite the complexity of the LSTM, highlighting that classical models can still be reliable and accurate in situations requiring high interpretability or limited data. On the other hand, recent research demonstrated the opposite. Badal et al. (2023)[31] evaluated feed-forward ANN, GARCH, and ARIMA for tomato price forecasting in the Indian markets, and the neural network obtained the highest accuracy. These contradictory results imply that although contemporary ML frequently performs well, the relative effectiveness of conventional versus sophisticated models may vary depending on the data properties and the commodity price behaviour. Even as the field moves towards non-linear approaches, ARIMA and related time-series models are still commonly used for benchmarking and serve as a helpful baseline overall.

Machine learning and deep learning approaches

Researchers have increasingly used ML and deep learning (DL) techniques to improve forecasting performance for perishable commodities, driven by the growing availability of price data and computing power. ML/DL techniques, unlike parametric time-series models, can identify multivariate dependencies and non-linear patterns in the data. Recent findings also show that hybrid ensemble learning frameworks that use a single meta-learning system to combine multiple ensemble strategies, such as bagging, boosting, stacking, and dagging, can achieve more accurate and stable predictions for highly unstable financial time series[32]. However, in practice, there hasn't been much use of traditional machine learning algorithms (such as gradient boosting, random forests, or support vector regression) for predicting fruit prices. Recent literature suggests that very few studies have examined advanced ML specifically for fruits or vegetables, and the majority of price-prediction studies for fresh produce still use basic models (ARIMA, simple ANN, SVM) [33–37]. However, Zhang et al.[38] presented a LightGBM boosting model for fresh agricultural prices (including bananas) and reported better accuracy and stability than ARIMA, SVM, and feed-forward neural nets, primarily based on an ensemble tree-based approach. LightGBM demonstrated the promise of contemporary ML by achieving the lowest error and proving greater resilience as the forecast horizon increased in their evaluation across several commodities.

The ability of DL techniques to automatically recognize intricate time-series patterns has made them popular. Forecasting agricultural prices has extensively used recurrent neural networks, particularly LSTM networks. Static models are unable to learn non-linear relationships and long-term dependencies[19]. Empirical studies[19] consistently show that LSTM-based models outperform traditional models for volatile crop prices across various error metrics. For instance, when predicting apple prices in the Indian market, Thakur et al.[28] discovered that an LSTM-RNN produced lower RMSE and MAPE than the top ARIMA and GARCH models. The advantage of LSTM in capturing nonlinear price-driving factors (weather shocks, supply lags, etc.) was demonstrated by its superior ability to fit the actual price trajectory and to generalize beyond the training data. According to Yin et

al. (2020)[39], researchers also found that proposed STL-LSTM networks reduce prediction error in the vegetable price data, particularly for abrupt price spikes. This is mainly because LSTM can model irregular jumps and seasonal effects in time series. With this method, accuracy increased significantly. For highly seasonal price series, MAPE dropped to about 7%, and the STL preprocessing and attention produced a prediction accuracy that was roughly 12% higher than a vanilla attention-LSTM without any decomposition.

There has also been an investigation into other deep architectures, such as for crop price forecasting. Murugesan et al. (2022)[40] assessed variants such as CNN-LSTM hybrids, stacked LSTM layers, bi-directional LSTM, and even ConvLSTM. These tests demonstrate how deep models can be customized to capture various aspects of price behavior, such as cycles, trends, and volatility, and frequently outperform individual traditional or machine learning models in terms of predictive performance. Although RNN/LSTM variants are still the most widely used in fruit price applications, convolutional neural networks and hybrid deep networks have also been tested alongside recurrent models[41–43]. Using data on vegetable prices, Wu and Sun (2025)[26] assessed state-of-the-art models (including more recent Transformers such as PatchTST and "iTransformer") and found that these DL models performed well overall. Sari et al. (2024)[44] used a neural network optimised by a genetic algorithm for Extreme Learning Machines (ELMs) to forecast the prices of 11 major agricultural commodities. Their ELM-GA model outperformed both a comparator LSTM-GA model and a benchmark ARIMA across all 11 commodities, even during the high volatility of 2022's price swings. In complex, real-world datasets, ML techniques (such as optimised ELMs or hybrid networks) can outperform both conventional models and conventional deep networks. This study was one of the first to predict a large set of commodities over extended periods successfully.

Case studies on various individual fruits further illustrate the benefits of advanced models. Ma et al. (2023)[44] used an LSTM model and a back-propagation neural network to forecast daily mango prices with remarkably high accuracy. In testing, the hybrid BP-LSTM model outperformed a standalone BP network and a single LSTM, achieving an MAPE below 0.2% and an R^2 of 0.99. These findings demonstrated that meticulously crafted DL models (and their ensembles) can fit the real price trajectories of highly volatile, seasonal fruits with previously unheard-of accuracy. Recent studies also highlight the significance of tuning and model generalisation. Nayak et al. (2025)[4] presented a meta-transformer framework that combines Transformers with metaheuristic algorithms for hyperparameter optimisation. Their Transformer models (with PSO/GWO/WOA optimisers) produced 70–90% lower error than a GARCH benchmark when applied to potato prices. This is part of a larger movement to improve forecasting accuracy and stability by combining cutting-edge DL architectures with optimisation methods.

Real-time forecasting and deployment frameworks

The shift to real-time forecasting and deployable systems is an equally significant component of related work. Still, the majority of the studies mentioned previously focus on modeling accuracy using historical data. Researchers have proposed hybrid architectures combining machine learning models with classical time-series components and stochastic simulation to predict agricultural commodity prices. These models would link forecasting performance with risk analysis and food security concerns at both the national and regional levels[45]. Forecasts of fruit prices are most helpful in practice when they are promptly provided to farmers, traders, and policymakers so they can make informed decisions on storage, market timing, and policy interventions. This has also been recognized by the recent studies as well, primarily based upon the fact that the forecasting frameworks' scalability and latency are essential in addition to the accuracy[4]. Deployment architectures that can continuously update forecasts and ingest real-time data, such as daily market arrivals and prices, are gaining high popularity among researchers and practitioners. Some studies have also suggested pipeline designs in which cloud-based predictive models receive fresh price data and produce forecasts that farmers can view on a user dashboard or mobile app. It can be challenging to guarantee low latency in such systems, particularly because deep ensembles and transformer models can be computationally demanding.

Outside academia, the push for real-time price forecasting is evident in industry initiatives and government programs. Large agribusiness companies and retailers have begun investing in AI-driven price prediction systems. Notably, Walmart announced trials of a ML price-forecasting tool to predict the future prices and availability of fresh produce and other staples[46]. The objective is to anticipate price fluctuations to optimize inventory and procurement, which is a clear indication that real-time forecasting is transitioning from research to operational practice. Digital agriculture platforms and agri-tech startups also offer forecasting services. To bridge the gap between sophisticated predictive analytics and real-world farm-level decision-making, evolving platforms in India, such as StarAgri and Cropway, provide AI-driven advisory services, offer real-time commodity price forecasts to assist farmers in choosing which crop to plant or when to sell[47,48]. These deployment frameworks often integrate a user-friendly interface (mobile apps, SMS alerts) with back-end predictive models that are updated daily or weekly.

Recent research in this area has not only enhanced modelling techniques but also increasingly considered model deployment, particularly towards delivering accurate forecasts in a timely, usable format. At the macro level, adaptive food price forecasting systems are now being deployed to improve public information and guide policy responses during periods of rapid economic change, underscoring the importance of operational, uncertainty-aware forecasts for decision-makers[49]. However, there are still problems in this area, such as how well the model performs, how easy it is for end users to understand, how good the data is, and how well it fits. Contemporary research is aiding the development of prototype systems that demonstrate how an automated system could deliver an effective forecast, such as predicting cherry prices in the near future. Besides, these frameworks demonstrate how model updating and streaming data-handling methods have improved over time. Also, recent research indicates that enhancing forecast accuracy is merely the initial step; the subsequent step is ensuring that these forecasts are applicable in real time to support effective financial decisions in perishable

sectors. In India, though, platforms like e-NAM or private agri-tech advisories mostly share static analytics or observed market prices and not AI-enabled predictive forecasts. Moreover, there is a dearth of studies that have integrated advanced DL architectures with prospective, real-time implementation during an operational horticultural season. Consequently, this study is a unique effort to implement and validate a forecasting system in the Himalayan states of India, particularly in perishable fruit markets.

Materials and methods

This section presents the end-to-end methodological framework adopted for forecasting cherry prices using AI/ML models. It includes data collection, preprocessing, model training, evaluation, and deployment strategies.

Study context and market selection

To model and predict price dynamics in the regional cherry value chain, the current study was conducted across important cherry markets in Jammu and Kashmir (J&K), India, and a primary external wholesale market in Delhi (the national capital). The markets included in the study are presented in Fig. 1 and cover Azadpur Delhi, Parimpore Srinagar, Zazna Ganderbal, Aglar Shopian, and Narwal Jammu. The market selection was primarily governed by contributions to overall price discovery mechanisms for cherry commodities, both within and outside the J&K region, as well as by their role in regional trade volumes and spatial linkages. During the marketing season, the main hubs for the aggregation and exchange of cherry produce in the Kashmir Valley are the horticultural clusters of Parimpore Srinagar, Zazna Ganderbal, and Aglar Shopian. To facilitate inter-regional trade flows between Kashmir and the rest of the nation, Narwal Jammu serves as an intermediary nodal market and accounts for inter-state demand dynamics, price transmission behavior, and external market signals influencing cherry prices in J&K.

Regarding varietal considerations, the study focused on the cherry variety Makhmali (Guigne Noir Hative), which is widely grown across Jammu and Kashmir's major cherry-producing belts and accounts for a significant share of both market value and trade volume. To properly capture the diversity in supply chains and customer preference structures typical of large-scale inter-state markets, the analysis in the Azadpur market included both Makhmali and Mishri (Bigarreau Noir Grossa) varieties. Three well-known commercial quality grades, namely Super, Special, and Fancy, that are primarily determined by size, color, firmness, and general aesthetic parameters, have a significant impact on price formation and consumer preferences. Therefore, they were included in the data collection based on market segmentation and price differentiation analysis.

Data collection and curation

The daily wholesale cherry price data were obtained directly from official market records maintained by the local horticultural boards and Agricultural Produce Market Committees (APMCs) in each study market. The research team made several visits to the market across different marketing seasons to ensure they had full access and coverage. They worked closely with multiple stakeholders, including market secretaries, licensed traders, commission agents, and auctioneers, to collect data. The collected data was digitized and exported to Microsoft Excel spreadsheets for analysis. Moreover, during data handling, certain discrepancies have been observed, primarily due to variation in reporting formats across APMCs and the lack of records from the off-season. To address these problems, data preprocessing was performed. This ensured that model training and evaluation used only consistent, high-quality data. For the authenticity and data accuracy, a double-entry validation protocol was used. Any differences during the process were cross-checked with the original registered data.

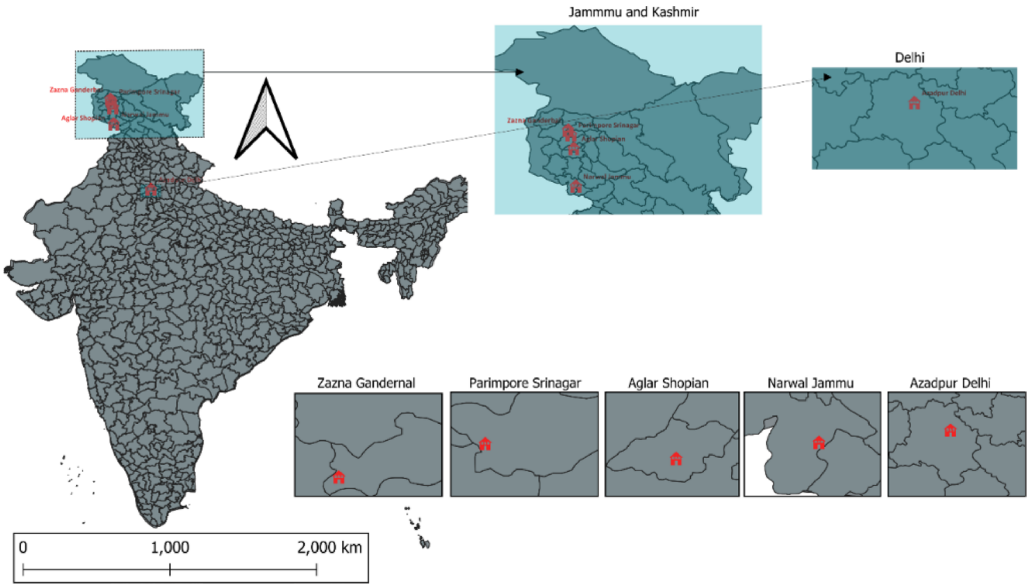


Fig. 1. Study area and major markets receiving cherry produce from Jammu & Kashmir.

Also, random spot checks were conducted to reduce transcription and encoding errors. Though due care and caution were taken in handling and analyzing the data, the possibility of bias may still exist in the collected data beyond the authors' control.

Data architecture: temporal scope and sale period segmentation

The data used in this study included daily cherry produce price observations gathered from the chosen markets over a range of time periods, contingent on the commercial significance of the cherry trade in each area and the accessibility of official records. Historical records span 2015–2024 for Makhmali and 2012–2024 for Mishri at Azadpur Market, 2019–2024 for Ganderbal, Parimpore Srinagar, and Aglar Shopian in the Kashmir valley, and 2016–2024 for Narwal Jammu. These differences in temporal coverage were due to the registers not being consistently archived across markets, and only the years for which files were accessible during field visits had their datasets digitised. Real-time forecasting for the 2025 marketing season was conducted after models were developed and validated using these historical datasets, enabling a prospective comparison of model performance against observed market prices.

The daily frequency at which all datasets were recorded reflects the level of detail at which cherry prices are reported and exchanged in wholesale markets throughout the marketing season. The analysis was explicitly limited to the specified sale periods of each year, which corresponded to active cherry trading in the respective markets due to the highly seasonal nature of cherry cultivation and market activity. The agro-climatic factors affecting cherry ripening, harvest schedules, and market arrivals were taken into consideration when determining these sale times. Specifically, the Mishri variety was sold later in the season (10 June–15 July in Azadpur) than Makhmali (20 May–30 June in Azadpur; May–June windows in valley markets). As an inter-state nodal hub, Narwal (Jammu) had the longest trading window among regional markets (15 May–10 July). Table 1 offers comprehensive temporal coverage of grades, varieties, and sale times for every market.

The study's methodological approach was to limit model evaluation and training to the specified sales periods for each year. By avoiding confounding effects from including off-season data that do not accurately reflect cherry market activity, this method ensured that models were calibrated and tested under real-world trading conditions and seasonal price dynamics. The study sought to improve the practical relevance, interpretability, and predictive accuracy of forecasting models by aligning the analytical timeframes with the commercially and biologically significant cherry production and marketing periods.

Data engineering and transformation pipeline

To ensure accuracy, consistency, and statistical validity, datasets were methodically pre-processed before model training. Occasionally, the resulting daily time series contained missing entries, transcription errors, and scale inconsistencies because the original market registers were kept on paper before digitisation. Thus, preprocessing was developed to enhance the numerical stability of machine learning and deep learning architectures (e.g., gradient-based training scaling) and to meet the statistical assumptions of traditional time-series models (e.g., stationarity for SARIMA). Imputation of missing values, stationarity diagnostics and differencing when required, and normalisation of inputs for neural models comprised the three primary steps of the preprocessing pipeline.

Missing-value imputation

A systematic three-step imputation method was used to resolve these discrepancies. The forward-fill (ffill) method was initially used to fill in missing values by carrying forward the most recent valid observation. A backward-fill (bfill) was used wherever a forward-fill was not feasible, such as when a missing entry occurred at the start of the series. In the rare instances where three or more consecutive values were missing, linear interpolation was used. Specifically, the missing value at time t was estimated based on the known values y_{t-k} and y_{t+k} as:

$$\widetilde{y_t} = y_{t-k} + \left(\frac{y_{t+k} - y_{t-k}}{2k}\right) \cdot (t - (t-k))$$

Assuming that the gaps surrounding the missing period are symmetrical, this combined strategy minimized artificial distortions brought on by data gaps while preserving local market fluctuations.

Market	Variety	Grades	Data period	Sale period
Azadpur Delhi	Makhmali	Super, Special, Fancy	2015–2024	May 20–June 30
	Mishri	Super, Special, Fancy	2012–2024	June 10–July 15
Zazna Ganderbal	Makhmali	Super, Special, Fancy	2019–2024	May 13–June 28
Narwal Jammu	Makhmali	Super, Special, Fancy	2016–2024	May 15–July 10
Parimpore Srinagar	Makhmali	Super, Special, Fancy	2019–2024	May 10–June 30
Aglar Shopian	Makhmali	Super, Special, Fancy	2019–2024	May 20–July 10

Table 1. Temporal coverage, varieties, grades, and sale periods of cherry price data across study markets.

Stationarity diagnostics and differencing
All price series underwent formal stationarity testing after imputation. This was necessary to meet the assumptions of conventional time-series models, such as SARIMA, which require stationary input data to generate accurate forecasts. For this, the Augmented Dickey-Fuller (ADF) test was used.

$$\Delta y_t = \alpha + \beta_t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-1} + \epsilon_t$$

where Δy_t is the first difference of the series, α is a constant, β_t captures any deterministic time trend, γ is the lagged level term's coefficient, and the summation uses lagged differenced terms to account for higher-order autocorrelation. The null hypothesis is accepted, i.e., $H_0 : \gamma = 0$, when a unit root is present, suggesting non-stationarity. Moreover, the series is said to be stationary if this null is rejected. To guarantee model parsimony, the Akaike Information Criterion (AIC) was used to calculate the lag length p. The series was differenced appropriately, both non-seasonally and seasonally when required, until stationarity was attained in cases where the unit root was present. The SARIMA model structure was then updated to include matching differencing orders (d, D). Strict stationarity was not required for ML & DL models; however, different versions were also provided to enhance convergence stability and lower the possibility of erroneous non-stationary patterns. To guarantee numerical stability in gradient-based optimisation, all price series were normalised before training the LSTM and Transformer models. The deployed pipeline used the Min–Max Scaler (range [0, 1]), and robustness checks also tested the Standard Scaler (zero mean, unit variance).

The MinMax scaling was defined as:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

which rescales observations into the interval [0, 1].

The Standard scaling was defined as:

$$x' = \frac{x - \mu}{\sigma}$$

where μ is the sample mean and σ the standard deviation. This preprocessing step enabled DL models to converge more quickly and steadily by preventing high-magnitude values from dominating the learning dynamics.

Forecasting models and theoretical formulations

Modeling was performed using a variety of techniques, from statistical baselines to advanced DL architectures, to predict cherry prices across selected markets with high accuracy. A realistic depiction of price discovery conditions was enabled by training the models on active trading windows determined by empirical sale periods. A crucial part of the study was the modeling phase, which allowed for both prospective real-time deployment and retrospective validation.

Seasonal autoregressive integrated moving average (SARIMA)
An extension of the ARIMA model, SARIMA accounts for both seasonal and non-seasonal components in time series data, making it especially well-suited to capturing strong intra-annual seasonality. Each univariate SARIMA model was specified as SARIMA (p, d, q) (P, D, Q) [s], where (p, d, q) stand for the non-seasonal orders of moving average (MA), autoregression (AR), and differencing (I), and (P, D, Q) stands for the corresponding seasonal components. The seasonal period (e.g., 7 for weekly and 30 for monthly periodicity) is denoted by s. Thus, the SARIMA model has the following mathematical form:

$$\Phi_p\left(B^s\right)\phi_p\left(B\right)\left(1-B\right)^d\left(1-B^s\right)^D y_t = \Theta_Q(B^s)\theta_q(B)\varepsilon_t$$

where, B is the backshift operator, polynomials $\phi_p\left(B\right)$ and $\theta_q(B)$ are non-seasonal AR and MA components, respectively, while $\Phi_p\left(B^s\right)$ and $\Theta_Q(B^s)$ are their seasonal counterparts. The terms $\left(1-B\right)^d$ and $\left(1-B^s\right)^D$ denote non-seasonal and seasonal differencing to achieve stationarity.

To identify potential values for p, d, q, P, D, and Q, plots of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) were analyzed. The final model structure was chosen for each market-variety-grade combination based on the minimum Bayesian Information Criterion (BIC) and Akaike Information Criterion (AIC). A schematic workflow of the SARIMA model development process, including stationarity testing, differencing, parameter identification, model fitting, and evaluation, is presented in Fig. 2.

Prophet
The Facebook Prophet model was used as an additional baseline to SARIMA because it handled seasonality, holidays, and non-linear trends intuitively and required minor adjustment. Facebook research developed Prophet, a modular additive model that performs well on business time series with significant seasonal effects and missing data. As such, it can be used to model changes in cherry prices driven by regional shocks and annual harvest cycles. The Prophet model's overall framework is described as follows:

$$y\left(t\right) = g\left(t\right) + s\left(t\right) + h\left(t\right) + \epsilon_t$$

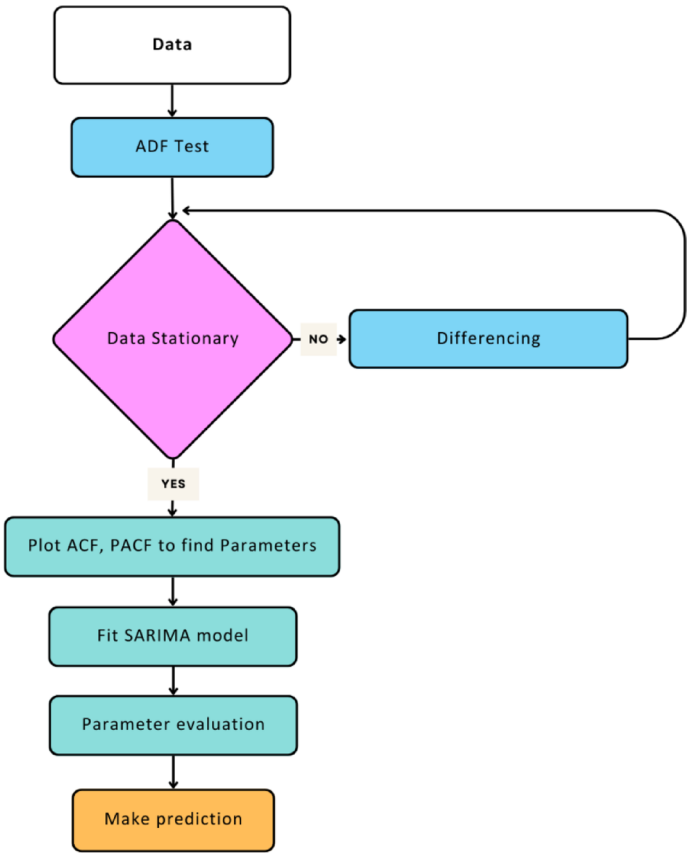


Fig. 2. Workflow outlining the key steps of SARIMA model specification and the forecasting procedure.

where y(t) is the observed value at time t, g(t) is the trend component, s(t) is the seasonal component, h(t) accounts for any holiday effects and ϵ_t represents the error term, accounting for noise and data irregularities.

Seasonality in the model captures periodic fluctuations, supporting both yearly and weekly seasonal effects, and is modeled using a Fourier series to represent the periodic nature of the data:

$$s\left(t\right)=\sum_{k=1}^{K}\left(a_k\cos\left(\frac{2\pi kt}{p}\right)+b_k\sin\left(\frac{2\pi kt}{p}\right)\right)$$

where K is the number of Fourier terms, p is the period of the seasonality, and a_k and b_k are the coefficients for the cosine and sine terms, respectively. The overall additive decomposition of Prophet into these four components is illustrated in Fig. 3.

Random forest (RF)

When dealing with tabular time series problems with moderate lags and where parametric modeling of autocorrelation and exogenous patterns is challenging, RF is efficient. Compared to conventional autoregressive models, it manages noise and missing values better and offers robustness against overfitting. The RF model consists of an ensemble of B regression trees. $\{T_b\}_{b=1}^{B}$, each trained on a bootstrap sample of the original dataset. The final prediction for a data point X is computed as the average of predictions across all trees:

$$\widehat{y}\left(x\right)=\frac{1}{B}\sum_{b=1}^{B}T_b(x)$$

A subset of predictors chosen at random from each node is used to recursively split the feature space to create each unique decision tree (T_b). The split criterion reduces the variance of the output variable within the resulting child nodes. This procedure guarantees low ensemble bias and decorrelation between trees.

In our model, the sale-period data from each market-year combination was used to construct all input features. By doing this, the model is guaranteed to learn from periods of economic activity, which are characterized by greater volatility, more abrupt price changes, and market-driven heterogeneity. By omitting off-season flatline periods, this approach also enhances generalization. The RF workflow, where multiple decision trees are trained on bootstrapped samples and their outputs are averaged to generate the final forecast, is illustrated in Fig. 4.

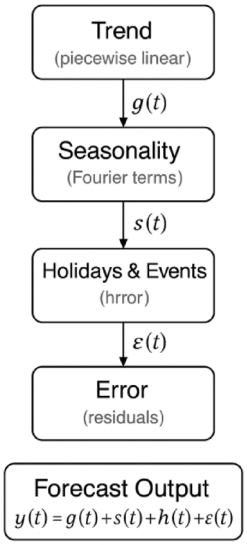


Fig. 3. Conceptual schematic of the Prophet model and its key components.

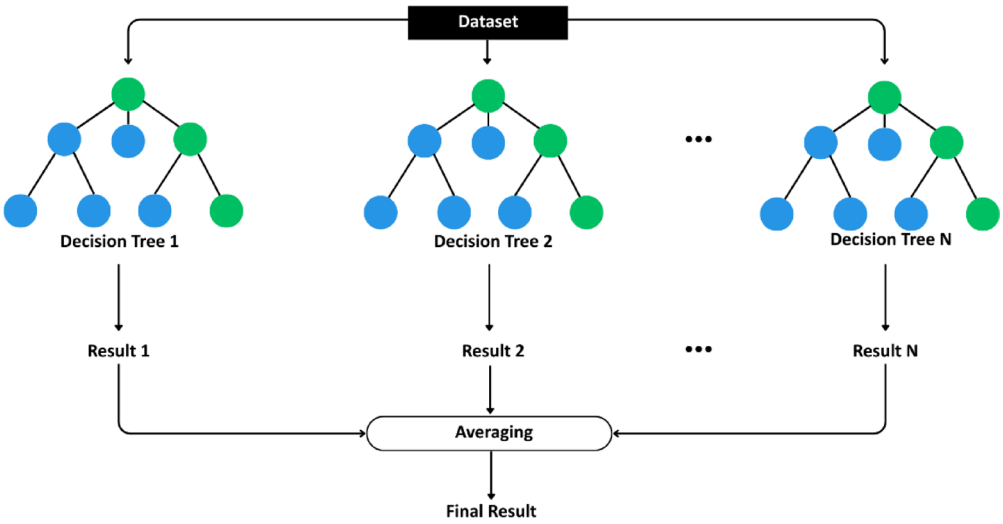


Fig. 4. Conceptual framework illustrating the structure and workflow of the RF model.

Extreme gradient boosting (XGBoost)
A scalable, high-performing gradient boosted decision tree (GBDT) implementation, XGBoost works best with structured datasets that have engineered features and moderate temporal horizons. It is an appealing option for complex forecasting tasks, such as those involving commodity market price dynamics, due to its integrated regularization, handling of missing values, and parallelized learning. In contrast to RF, which uses bagging to build trees independently, XGBoost sequentially builds trees (f_t), with each new tree attempting to lower the residual errors of the ensemble of earlier trees. Formally, the model prediction at iteration t is provided by:

$$\widehat{y}_i^{(t)} = \sum_{k=1}^{t} f_k\left(x_i\right), f_k \epsilon \mathcal{F}$$

where $\mathcal{F}$ denotes the space of regression trees. The objective function combines a differentiable loss function $\mathcal{L}$ and a regularization term Ω to penalize model complexity

$$\mathcal{L}^{(t)} = \sum_{i=1}^{n} l(y_i, \widehat{y}_i^{(t-1)} + f_t\left(x_i\right) + \Omega(f_t)$$

where,

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \; Not\ applicable$$

Here, T denotes the number of leaf nodes in a tree, w_j are the leaf weights, γ & λ are regularization parameters controlling the number of leaves and the magnitude of weights, respectively. The sequential boosting process, where each new tree is fitted to the residuals, is illustrated in Fig. 5.

Long short-term memory (LSTM)

It is a specific RNN architecture well-suited for time-series forecasting. It introduces gated mechanisms that allow selective retention, forgetting, & updating of information over arbitrary time lags., It is used for agricultural price forecasting, where both recent and delayed market events can influence price discovery, unlike conventional RNNs, which suffer from vanishing gradients and limited memory across long sequences.

The LSTM cell is governed by a series of non-linear transformations involving the input gate i_t, forget gate f_t, output gate o_t, candidate cell state $\widetilde{C}_t$, and the final cell state C_t. These are computed as:

$$\begin{aligned} f_t &= \sigma(W_f\,[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i\,[h_{t-1}, x_t] + b_i) \\ \widetilde{C}_t &= \tanh(W_C\,[h_{t-1}, x_t] + b_C) \\ O_t &= \sigma(W_o\,[h_{t-1}, x_t] + b_o) \\ h_t &= O_t \tanh\,(C_t) \end{aligned}$$

where x_t is the current input, h_{t-1} is the previous hidden state, σ the sigmoid function, and the W and b are weights and biases. This architecture allows LSTM to model long-term non-linear patterns in multivariate time series data with minimal prior feature engineering. The internal architecture of an LSTM cell and the interaction of these components are illustrated in Fig. 6.

Transformer

Transformer-based architectures, which have recently become popular in time series forecasting due to the ability to model intricate global patterns without the recurrence bottleneck that RNNs and LSTMs have, were used in the current study. Transformer's self-attention mechanism, which was first proposed for machine translation[23], enables simultaneous focus on all time steps in a sequence, making it a helpful framework for sequence-to-sequence learning in the financial and agricultural domains. A schematic of the standard Transformer encoder–decoder architecture, highlighting its multi-head attention and positional encoding components, is shown in Fig. 7.

The multi-head scaled dot-product self-attention, which computes an attention score for each pair of time steps in a sequence, is the central component of the Transformer. For a given input sequence of query Q, key K, and value V, the attention is computed as:

$$Attention\,(Q, K, V) = softmax\left(\frac{QK^t}{\sqrt{d_k}}\right)V$$

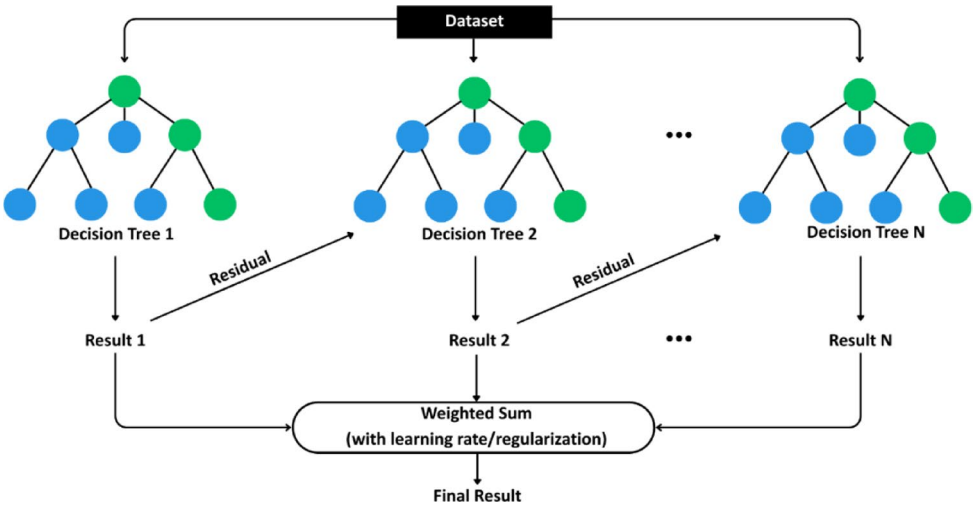


Fig. 5. Conceptual framework illustrating the structure and workflow of the XGBoost model.

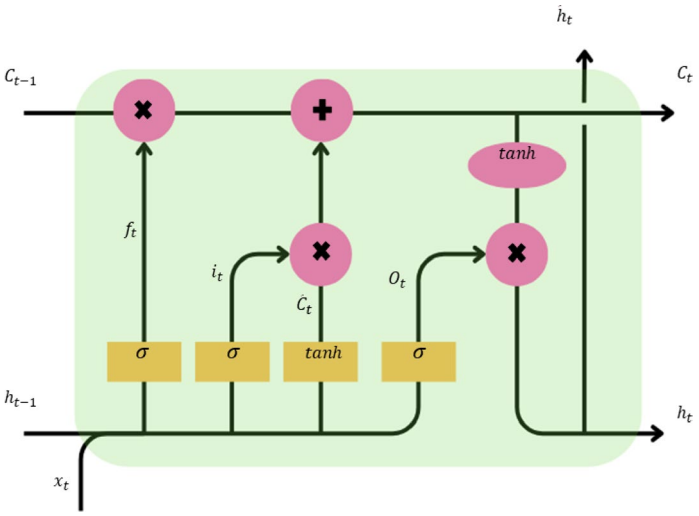


Fig. 6. Schematic representation of the internal structure of an LSTM cell.

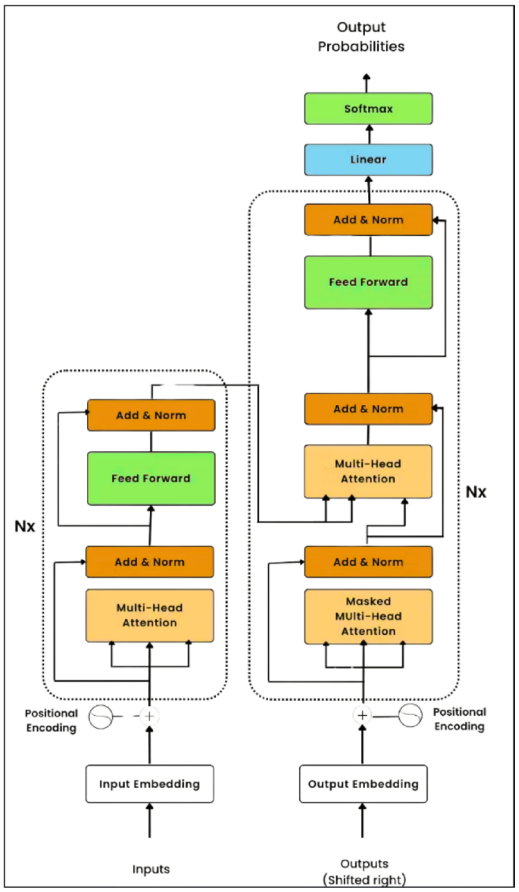


Fig. 7. Conceptual schematic illustrating the architecture and attention-based components of the transformer model.

where d_k, is the dimension of the key vectors. The multi-head attention mechanism projects the input into multiple spaces to capture diverse patterns:

$$MultiHead\,(Q,K,V) = Concat(head_1,\ldots,head_h)W^O$$

Each head is computed using separate parameter matrices, enhancing the model's ability to jointly attend to information at different temporal positions and feature scales. Since Transformer does not employ recurrence, positional encodings are appended to the input embeddings to maintain the temporal ordering:

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right), PE_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

These sinusoidal encodings enable the model to learn position-specific dependencies over time.

Model evaluation framework and error metrics

A wide range of statistical metrics was used to assess the predictive performance of the forecasting models. These consist of Directional Accuracy (DA), Mean Forecast Error (MFE), Mean Absolute Error (MAE), Symmetric Mean Absolute Percentage Error (sMAPE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Normalized Mean Bias Error (NMBE). Moreover, the DM test was used to formally assess whether variations in predictive performance were significant by statistically comparing forecast accuracy across rival models. Taken together, these metrics provide a strong picture of the model's bias, directional correctness, error magnitude, and accuracy. MAE measures the average magnitude of the forecast errors, treating all deviations equally regardless of direction. It is expressed as:

$$MAE = \frac{1}{n}\sum_{t=1}^{n}\left|Y_t - \widehat{Y}_t\right|$$

RMSE, derived from MSE, provides an interpretable measure in the same unit as the original prices and is calculated as:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(Y_t - \widehat{Y}_t\right)^2}$$

When actual or predicted values get close to zero, the sMAPE adjusts symmetrically to prevent bias and quantifies relative errors as a percentage:

$$sMAPE = \frac{100\%}{n}\sum_{t=1}^{n}\frac{\left|\widehat{Y}_t - Y_t\right|}{(|Y_t| + |\widehat{Y}_t|)/2}$$

This metric is handy for comparing forecast accuracy across multiple price ranges and markets. In comparison to the prior time step, DA evaluates the model's capacity to accurately forecast the direction of price movement, i.e., an increase or decrease. It is stated as follows:

$$DA = \frac{1}{n-1}\sum_{t=2}^{n}1[(Y_t - Y_{t-1})\left(\widehat{Y}_t - \widehat{Y}_{t-1}\right) > 0]$$

where 1 [·] is an indicator function that, if the sign of the actual & predicted movements is the same, equals 1, DA, which is expressed as a percentage, is critical when making decisions about market operations. MFE provides insight into systematic bias by capturing the average signed difference between actual and predicted values:

$$MFE = \frac{1}{n}\sum_{t=1}^{n}(Y_t - \widehat{Y}_t)$$

A negative MFE indicates underestimation, while a positive MFE indicates overestimation. By expressing the average forecast bias as a percentage of the mean observed value, the NMBE facilitates comparisons across price points:

$$NMBE = \frac{1}{n}\sum_{t=1}^{n}\frac{(Y_t - \widehat{Y}_t)}{\overline{Y}} \times 100\%$$

where $\overline{Y}$ is the mean of the observed values. NMBE helps understand relative bias and compare forecast calibration across varieties, markets, or time periods. The variables used in the metric formulas are Y_t, which

represents the actual observed values at time t, and $\widehat{Y}_t$, which denotes the predicted values generated by the forecasting model for the same time point.

Diebold Mariano (DM) test for forecast comparison

Suppose that we have two forecasts $f_1, \ldots, f_n$ & $g_1, \ldots, g_n$ for a time series if $y_1, \ldots, y_n$ and want to see which forecast is more accurate. Based on one of the error measurements, the approach is to select the forecast that has the smallest error measurement. The DM test is most suitable for determining whether the two forecasts are significantly different. Let e_i and r_i be the residuals for the two forecasts, i.e.

$$e_i = y_i - f_i r_i = y_i - g_i$$

and let d_i be defined as one of the following

$$d_i = e_i^2 - r_i^2$$

The time series d_i is called the loss-differential. Clearly, the first of these formulas is related to the MSE error statistic, and the second to the MAE error statistic. We now define

$$\overline{d} = \frac{1}{n}\sum_{i=1}^{n} d_i$$

For $n > k \geq 1$, define

$$\gamma_k = \frac{1}{n}\sum_{i=k+1}^{n} (d_i - \overline{d})(d_{i-k} - \overline{d})$$

γ_k is the auto-covariance at lag k.

For $h \geq 1$, define DM Statistic as follows:

$$DM = \frac{\overline{d}}{\sqrt{\frac{[\gamma_0 + 2\sum_{k=1}^{h-1}\gamma_k]}{n}}}$$

It is generally sufficient to use the value, $h = n^{\frac{1}{3}} + 1$, There is a significant difference between the forecasts if $|DM| > z_{crit.}$ where $z_{crit.}$ is the two-tailed critical value for the standard normal distribution. The key assumption for using the DM test is that the loss differential is a time series. d_i is stationary.

All models used in the study underwent the same training, validation, and test procedures to ensure comparability and reproducibility. The 2024 data were set aside for testing, while the training set included all observations for each market-variety-grade series from the beginning of the historical sales period through the penultimate year available, i.e., 2023 in the present case. The final 10% of observations in each training series was set aside as a validation part for early stopping and model selection. By avoiding look-ahead bias and mirroring the intended real-time forecasting scenario, this design ensured that the split points reflected the actual marketing windows for each variety and market.

To prevent over-tuning and ensure a fair comparison, model-specific settings were deliberately kept light and symmetrical. We ran a compact grid over (p,d,q,P,D,Q) $\in$ {0,1} for SARIMA, setting the seasonal period "s" to the dominant within-season periodicity inferred from the ACF of each series (daily data usually yielded a weekly cycle, s≈7). AIC on the training window selected the final specification, and no data from the test window was used. Prophet only trained during the training window and used its default additive trend/seasonality with automatic changepoints (yearly and weekly components enabled). To control variance, we constructed univariate lag features for tree ensembles and capped the look-back horizon; XGBoost and RF used up to 60 lags. Simple forward validation on the held-out validation slice was used to select the number of lags (minimizing validation MSE); the other hyperparameters (RF n_estimators = 100, XGB n_estimators = 100, learning_rate = 0.1, max_depth = 3, subsample = 1.0) were set to library defaults. The input sequence length was selected from {3, 4, 5} based on validation loss, and neural models were trained on standardized targets (StandardScaler fit on the training set). The LSTM employed batch size 32, a single hidden layer of about 100 units with Adam, up to 100 epochs, and early stopping (patience = 10, best weights restored). With embedding dim = 32, feed-forward dim = 128, four encoder layers, Adam at 1e-3, up to 100 epochs, and the same early-stopping protocol, the Transformer was a lightweight encoder. All models were trained on identically masked sale-period days, with identical chronological train/validation/test splits that matched the actual season of each market variety. No exogenous variables were used. Conventional k-fold cross-validation was not applied, as it is unsuitable for autocorrelated time-series data. Instead, chronological splits aligned with market-specific sale windows were used, with a final validation slice within the training set for model selection and early stopping.

To ensure the results can be reproduced, random seeds were set where libraries allow (e.g., sklearn random_state = 42; similar seeds for NumPy, TensorFlow, and PyTorch). The computational environment used Python 3.10.16 (Anaconda distribution) on Windows 10 (64-bit, Intel i7 CPU), with core libraries including NumPy

2.1.3, pandas 2.3.0, scikit-learn 1.7.0, statsmodels 0.14.5, Prophet 1.1.6, XGBoost 3.0.4, TensorFlow 2.19.0/Keras 2.10.0, and PyTorch 2.3.1. We used Matplotlib 3.10.3 and Seaborn 0.13.2 to create the visualisations.

Real-time forecasting deployment architecture

The model chosen for operational deployment was the one that performed the best overall across test datasets, as demonstrated by the lowest error scores (MAE, RMSE, and MSE), the highest directional accuracy, and statistically superior results verified by the DM test. The chosen model was incorporated into a lightweight web service architecture to ensure the forecasting system's real-time usability, scalability, and accessibility. The completed model was loaded into a Python-based FastAPI backend after being serialized in Keras format, primarily because of its asynchronous, high-performance features, which make it ideal for handling multiple API calls. Programmatic access to forecasts based on user-specified market, fruit, variety, grade, and date-range inputs is enabled by the API's exposure of forecasting services via HTTP endpoints. To ensure continuous availability and automatic scaling, this FastAPI application was containerized and deployed to Render, a fully managed cloud platform. Agile development and smooth forecasting logic updates were made possible by Render's integration with Git-based CI/CD pipelines and its capacity to manage web services without incurring server overheads.

JavaScript-based frontend components were used to integrate the deployed API into the official Market Intelligence Cell (MIC) website so that users and stakeholders could access the forecasts. Figure 8 provides an overview of the entire pipeline from raw data acquisition to real-time forecast deployment via a REST API. The frontend dynamically queries the FastAPI endpoint and uses specially designed charts to display model outputs based on the dynamics of chosen markets in Delhi and Jammu & Kashmir. Based on this, producers, traders, and policy planners have direct access to real-time, location-specific market intelligence for the studied commodity.

In addition, a dedicated market data collection interface was developed using FastAPI forms for authorized market officials. This form allowed entry of daily fruit prices during the sale season, which were stored in a PostgreSQL cloud database linked to the backend. These official entries were automatically time-stamped and validated for completeness. A scheduled backend process continuously queried this database and, upon detecting new entries, triggered model updates and forecasts using the latest available data. These forecasts were then published through the same API pipeline and rendered on the dashboard, ensuring the system's real-time responsiveness. This live feedback loop enabled the dynamic integration of field-entered data with deep learning forecast pipelines, creating a closed system in which new market intelligence directly improved predictive capacity. The overall architecture of this integrated system is illustrated in Fig. 9.

Results and discussion

Descriptive statistics and price variability

The descriptive analysis of cherry prices across five major wholesale markets in Table 2 reveals significant inter-market, inter-varietal, and intra-varietal (grade-wise) differences on both mean prices and price volatility. The highest mean prices were consistently observed at Azadpur market, Asia's largest wholesale fruit hub and a

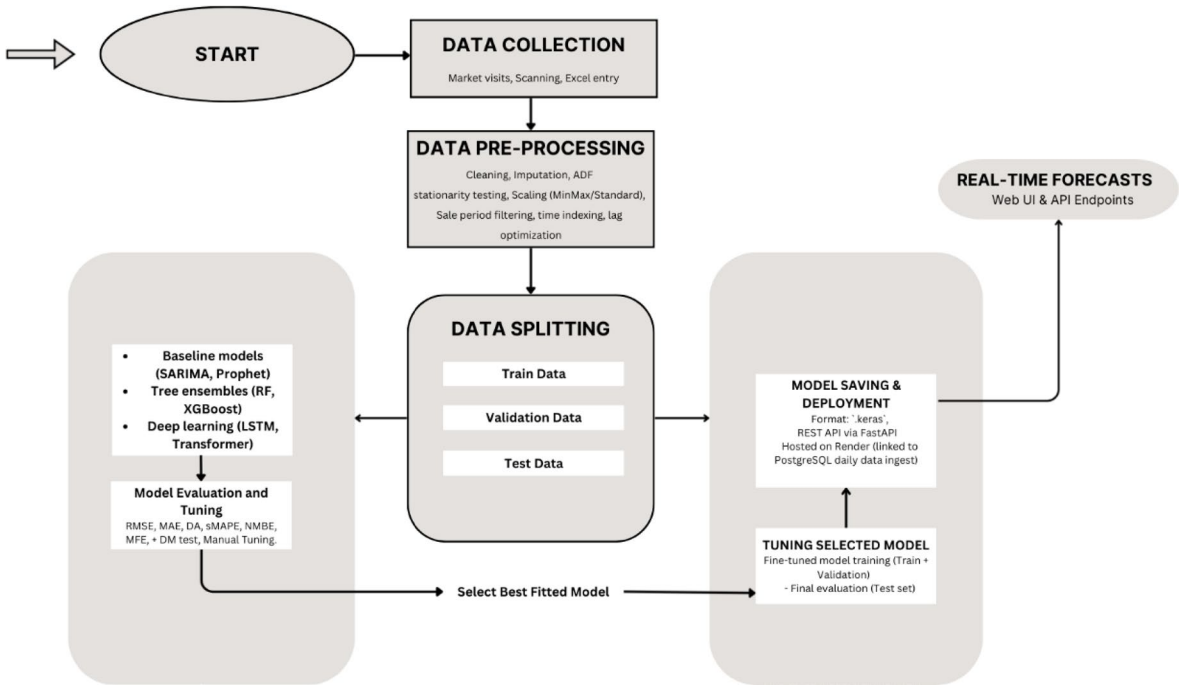


Fig. 8. End-to-end workflow depicting data collection, preprocessing, model selection, deployment, and real-time forecasting.

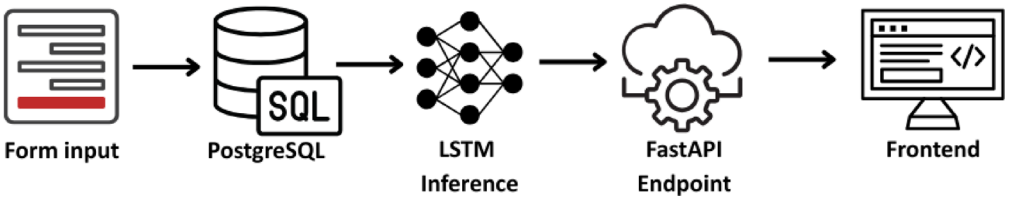


Fig. 9. Schematic architecture of the integrated forecasting system, from data input to model inference and frontend visualization.

Market	Variety	Grade	Mean ± SD (₹/kg)	C.V. (%)
Azadpur	Makhmali	Super	133.02 ± 51.37	38.62
		Special	96.99 ± 42.10	43.43
		Fancy	62.55 ± 24.66	39.45
	Misri	Super	179.55 ± 80.52	44.85
		Special	130.90 ± 57.84	44.20
		Fancy	83.46 ± 40.68	48.74
Ganderbal	Makhmali	Super	146.67 ± 33.48	22.82
		Special	100.92 ± 25.49	25.27
		Fancy	57.77 ± 12.20	21.11
Parimpore		Super	140.82 ± 36.90	26.20
		Special	104.98 ± 33.27	31.69
		Fancy	76.86 ± 25.72	33.46
Narwal		Super	141.95 ± 46.41	32.66
		Special	107.91 ± 44.95	41.65
		Fancy	70.34 ± 38.03	54.07
Shopian		Super	126.22 ± 48.77	38.64
		Special	90.48 ± 37.07	40.97
		Fancy	62.02 ± 26.29	42.40

Table 2. Mean prices and variability of cherry grades across markets and varieties (₹/kg).

significant re-export location. Misri cherries were significantly more expensive, the Super grade averaged at ₹179.55/kg (± ₹80.52), while Makhmali Super grade was sold at ₹133.02/kg (± ₹51.37). Misri's superior quality characteristics, such as firm texture, darker pigmentation, and longer shelf life, are likely reasons for the price premium, as they increase institutional and consumer demand in both distant and urban markets[50]. Nonetheless, Misri Super's coefficient of variation (C.V.) of 44.85% also shows significant price volatility, indicating sensitivity to supply and quality variations at Azadpur[43].

In Kashmir valley markets, particularly in Zazna Ganderbal, where Super grade cherries averaged at ₹146.67/kg with a comparatively low C.V. of 22.82%, the Makhmali variety showed more stable pricing, indicating consistent quality and localized trade dynamics (Table 1). On the other hand, despite being necessary aggregation or transit hubs, the Narwal and Shopian markets displayed more within-grade variability. The Narwal Fancy grade cherries had the highest overall C.V. (54.07%), indicating that lower-quality produce traded at Narwal is more susceptible to fluctuations. These patterns suggest that when supply volumes or quality of lower grades fluctuate, their prices swing more dramatically, a phenomenon also noted in other fruit markets where seasonal gluts and uneven quality destabilize prices[51]. In Parimpore, the Super grade averaged at ₹140.82/kg (C.V. = 26.20%), while Fancy sold at ₹76.86/kg with a higher C.V. of 33.46%. Shopian displayed similar patterns, with slightly lower mean prices and C.V. values around 38–42% for all three grades, reflecting greater intra-grade variability, particularly in the lower-quality segments. The Super grade consistently achieved the highest mean prices across all markets, followed by Special and Fancy, confirming the value of grading from a business standpoint[52,53].

Nonetheless, there were notable differences in price dispersion across markets and grades, with Fancy and Special grades typically exhibiting greater relative variation. This implies that while higher grades maintain more stable pricing, particularly in upstream markets like Azadpur and Ganderbal, lower grades are more vulnerable to the market-induced volatility. Similar observations have been reported in China[54] on apple and grape markets, where both showed significant price fluctuations over a decade of data, with grapes in particular exhibiting a high-risk, high-return volatility profile compared to the more stable apple prices. Typical fruit price volatility can be on the order of 20–25% C.V. in well-supplied markets[55], so the 40–50% C.V. observed for lower-grade cherries in some markets here signals exceptionally high instability. The fact that the most central market (Azadpur) and the top grades show lower relative variability is consistent with the idea that broader market access and better-

quality control help stabilize prices. In contrast, peripheral markets and lower grades face disproportionately larger swings[53,56].

Temporal dynamics and seasonal trends

The intra-seasonal swings and grade-specific stratification observed in actual trading environments are not adequately captured by descriptive statistics, despite their ability to provide helpful aggregate measures of mean prices and variability. To overcome this, we examined temporal price dynamics across markets and grades to better understand how volatility develops during the marketing season. These daily patterns are depicted in Fig. 10, which emphasises sharp intra-seasonal fluctuations, especially in peripheral markets, as well as the durability of grade-wise price tiers. A study on citrus fruits in Nepal found that prices are lowest right after harvest and then rise to a peak before the next harvest, reflecting strong seasonal production patterns[57]. The pricing of the Makhmali vs. Misri varieties in Azadpur Market was evidently tiered, with Super grades regularly fetching the highest prices, followed by Special and Fancy. With Super grade prices peaking above ₹350/kg during peak trading periods, the Misri variety in particular showed noticeable price spikes at several intervals, suggesting periods of high demand and potentially reduced supply from origin markets. Price trends in Ganderbal and Parimpore markets were comparatively smoother, with the former showing less volatility and a more distinct division between grades over the trading period. Compared with the Azadpur market, Ganderbal's Super grade prices maintained a narrower variance, occasionally reaching ₹200/kg.

Market-specific differences in volatility observed in cherry trading are likewise reported in broader agricultural markets. In general, large terminal markets tend to have more stable prices, whereas local or aggregation markets experience sharper fluctuations. A multi-market analysis of Indian apple trade revealed that price variation was lowest in Delhi's Azadpur terminal market (approx. 6%) and much higher in a producing-area market like Shimla/Dhalli (approx. 17%)[58]. Delhi's market stability was attributed to its central hub status for the arrival of fruits from different states and for imports. In contrast, primary assembly markets with fewer supply sources experience frequent shocks and short-term price jumps, as in the Shopian and Narwal markets, where prices rise and fall when local harvest volume fluctuates. These differences indicate that larger, well-integrated markets can better absorb supply shocks, while smaller regional markets are more susceptible to local supply–demand imbalances, corroborating the role of demand elasticity and market integration in price stability.

Stationarity and autocorrelation diagnostics

There are notable variations in the statistical characteristics of cherry price dynamics, as only 9 of the 18-time series analysed demonstrated stationarity at the 5 percent significance level (Table 3). While some mid and lower grade price series in peripheral markets showed strong evidence of mean-reverting behaviour (Narwal Fancy, ADF = −3.79, $p = 0.0027$), higher-grade price series like Azadpur Makhmali Super (ADF = − 1.03, $p = 0.74$) and Misri Super (ADF =− 2.53, $p = 0.11$) remain non-stationary. These results show that while localised markets show more short-term stabilisation, premium series typically endure long-term shocks and trends. A comparative summary of the ADF test results by market and grade is shown in Fig. 11.



Fig. 10. Time-series trends of average cherry prices (₹/kg) across multiple markets for different grades.

Market	Variety	Grade	ADF statistic	p-value
Azadpur	Makhmali	Super	-1.0309	0.7418
		Special	-1.4508	0.5557
		Fancy	-2.3617	0.1528
	Misri	Super	-2.5256	0.1093
		Special	-2.8074	0.0577
		Fancy	-3.1247	0.0247
Ganderbal	Makhmali	Super	-3.054	0.031
		Special	-3.9741	0.0016
		Fancy	-3.3339	0.0134
Parimpore		Super	-2.6165	0.0896
		Special	-2.579	0.0974
		Fancy	-2.905	0.0447
Narwal		Super	-3.6207	0.0054
		Special	-3.5471	0.0069
		Fancy	-3.7938	0.0027
Shopian		Super	-2.6478	0.0539
		Special	-3.9753	0.0015
		Fancy	-2.8236	0.0549

Table 3. ADF test results for the cherry price series across markets.

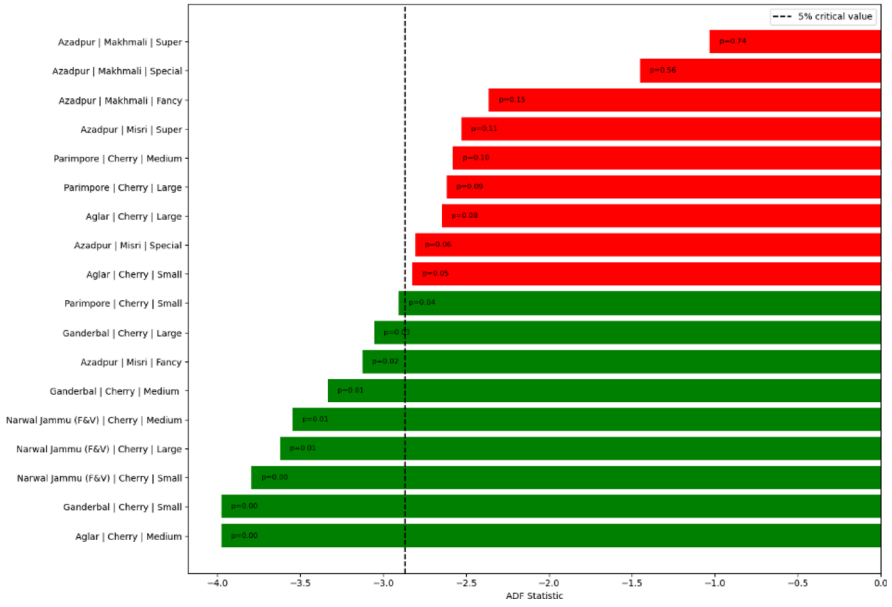


Fig. 11. ADF-based stationarity assessment of cherry price time series across different market and grade combinations.

To describe temporal dependencies and direct the identification of SARIMA order, autocorrelation and partial autocorrelation functions (ACF/PACF) were examined following the stationarity assessment (Figs. 12 and 13). Strong short- to medium-term persistence in price movements was indicated by the ACF plots for Azadpur and Parimpore, which showed slow, statistically significant decay over several lags. This behavior aligns with findings from Wani et al. (2021)[59], which reported similar ACF decay in apple prices across Indian markets, attributing the trend to predictable seasonal inflows and gradual market adjustments. In contrast, Ganderbal and Narwal's PACF plots revealed abrupt cut-offs following lags 1 or 2, indicating that AR(1) or AR(2) processes were dominant, which have previously been exploited for parsimonious SARIMA modeling in perishable commodities like grapes and mangoes[60–62]. Also, Shopian and Azadpur showed small but persistent spikes after lag 5, suggesting latent seasonality or localised shocks. All of these diagnostics point to a structural difference, smaller regional markets exhibit simpler autoregressive patterns and quicker adjustments, whereas persistent dynamics and prolonged memory distinguish larger terminal markets.

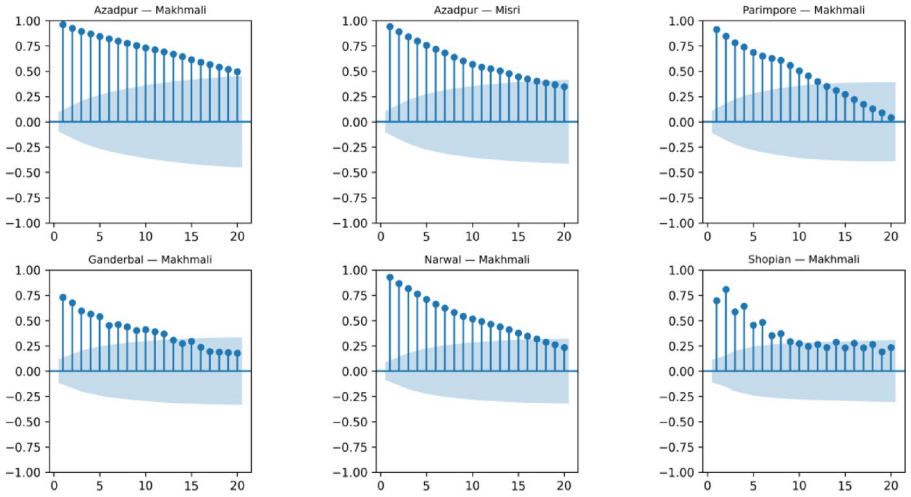


Fig. 12. Autocorrelation (ACF) plots of daily cherry price series across markets.

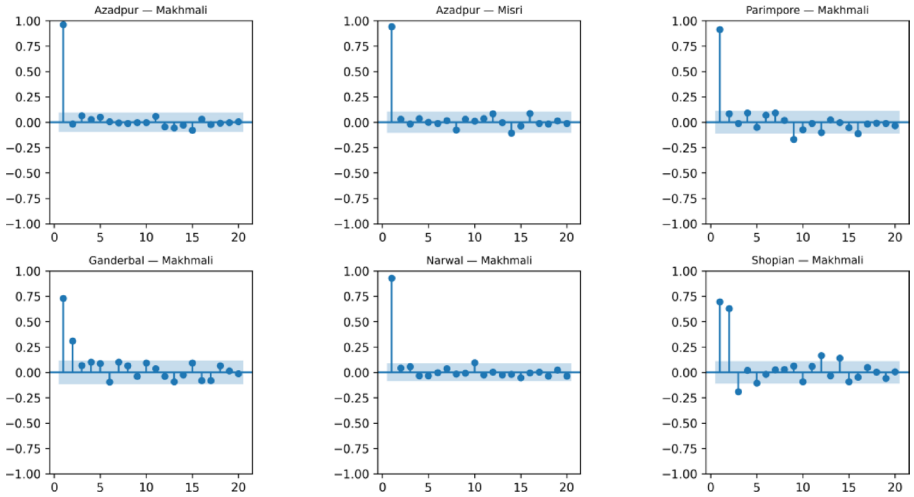


Fig. 13. Partial autocorrelation (PACF) plots of daily cherry price series across markets.

The prevalence of non-stationarity in premium grades likely reflects structural persistence driven by institutional demand and seasonal disruptions, similar to the patterns reported for tomato and onion markets in India[63,64]. Such behavior underscores the limitations of classical models like ARIMA when applied to agri-price data with embedded shocks or trend components. Conversely, the localized mean-reverting tendencies observed in some mid- and lower-grade series suggest that smaller, less commercially integrated markets can achieve short-term stability. As emphasized by various studies[65–67], ML and DL models, such as LSTM, are better suited for capturing dynamic and nonlinear patterns in horticultural prices, especially when faced with regime shifts and sparse data.

Model performance evaluation and robustness checks

Table 4 reports the top-performing forecasting model for each market–variety–grade combination, selected based on accuracy (MAE, RMSE, sMAPE), bias (MFE, NMBE), and DA, showing that DL methods, in particular LSTM and Transformer, consistently outperformed other methods in terms of accuracy and reliability. This finding is consistent with recent studies on commodity price forecasting, which reported that DL networks (e.g., LSTM, GRU) capture complex temporal patterns far better than ARIMA or linear models[19,68,69]. The present results showed an exceptionally low sMAPE of 2.74% in the Makhmali-Fancy segment at Azadpur using the LSTM model, setting a new benchmark for forecasting accuracy in premium grades. Also, Transformer models frequently have very low sMAPE values, typically from 3 to 9%, and DA scores of over 75% in a variety of grade-market combinations. For stakeholders to predict price trends rather than just absolute values, such high directional fidelity is essential, and advanced DL-based architectures have similarly excelled at predicting price movement direction in prior work[70].

Market	Variety	Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Azadpur	Makhmali	Super	LSTM	9.50	14.64	3.74	54.29	2.24	0.89
		Special	LSTM	11.41	19.61	6.00	48.57	-6.44	-3.19
		Fancy	LSTM	3.21	5.02	2.74	42.86	0.12	0.10
	Misri	Super	LSTM	21.54	22.7	7.05	51.61	-21.54	-6.73
		Special	Transformer	8.04	10.4	3.34	83.87	-5.89	-2.48
		Fancy	LSTM	12.29	19.77	8.08	51.61	-7.07	-4.61
Narwal	Makhmali	Super	Transformer	5.58	8.00	3.83	78.05	0.09	0.06
		Special	LSTM	8.64	12.96	9.38	48.78	0.23	0.24
		Fancy	Transformer	7.07	7.82	11.92	78.05	-3.32	-5.61
Ganderbal		Super	Transformer	5.58	8.00	3.83	78.05	0.09	0.06
		Special	Transformer	10.91	13.2	11.95	85.37	-1.28	-1.33
		Fancy	Transformer	7.07	7.82	11.92	78.05	-3.32	-5.61
Parimpore		Super	Transformer	12.71	14.33	10.68	93.75	1.98	1.54
		Special	LSTM	5.67	11.75	6.45	41.3	3.69	4.05
		Fancy	Transformer	5.61	8.68	9.36	65.96	5.61	8.58
Shopian		Super	Transformer	12.66	17.68	8.60	54.35	2.25	1.59
		Special	LSTM	7.54	19.23	7.70	62.00	-2.80	-3.03
		Fancy	Transformer	4.82	11.19	9.88	75.oo	1.06	2.20

Table 4. Top performing forecasting models by market, variety, and grade.

Model	Significant wins ($p < 0.05$)	Total pairwise tests	% of Wins	Avg. DM Stat (Wins)
LSTM	61	90	67.8%	−4.84
Transformer	48	90	53.3%	+3.85
RF	24	90	26.7%	−1.24
XGB	20	90	22.2%	+4.06
SARIMA	11	90	12.2%	+2.47
Prophet	5	90	5.6%	−6.58

Table 5. Summary of DM test results comparing forecasting models.

Importantly, both the MFE and the NMBE are near zero or have modest magnitudes in the majority of the top-performing configurations. This shows that the models are not consistently over- or under-predicting, an essential quality for ensuring accurate and valuable market insights. For example, Transformer models frequently achieve NMBE values within ± 5, demonstrating their flexibility in handling diverse data distributions with negligible drift. This lack of systematic bias aligns with best practices from other forecasting studies, where top models exhibit minimal bias alongside high accuracy[19]. Also, top models' having low forecast variance is confirmed by the MAE and RMSE metrics. Despite the inherent volatility of cherry markets, LSTM and Transformer models frequently report MAE values below 10 and RMSEs comfortably below 20 across several grade-variety segments. Such low error magnitudes in a volatile setting underscore the value of these DL models as decision-ready forecasting engines. Indeed, prior research has shown that deep networks can dramatically reduce forecast error in volatile commodity series, as in the study conducted by Manogna et al. (2025)[19], where a GRU network reduced RMSE by over 70% compared to ARIMA in forecasting onion prices. Such performance demonstrates their value as applicable, real-time forecasting frameworks and analytical tools.

Point-wise accuracy metrics (Table 4) show that Transformer and LSTM models perform better than others, with Transformer often the best model across particular market, variety, and grade segments. However, a formal DM test was used to see if these improvements were statistically significant. Table 5 summarizes the pairwise DM comparisons for each model. The results showed that LSTM had the most wins (61 out of 90 tests, or about 68%), which shows that it is strong across a wide range of market-grade combinations. Transformer also did well, especially in some volatile markets, winning more than half of its pairwise tests (53.3%). The difference between Tables 4 and 5 can be explained by the Transformer achieving better pointwise accuracy in some cases. Still, the LSTM consistently beat its competitors across the system, making it a better choice for deployment. On the other hand, the tree-based models (RF and XGBoost) had fewer significant wins, indicating they don't perform well against benchmarks in unstable pricing environments. Traditional models like SARIMA and Prophet had the fewest notable wins, suggesting they aren't very good at handling the complex dynamics of cherry prices. In conclusion, the DM test shows that the apparent superiority of DL models, especially LSTM, is statistically significant across many forecasting situations and is not just an artefact of error metrics. All detailed pairwise DM test outputs by market–variety–grade are provided in the supplementary materials[71].

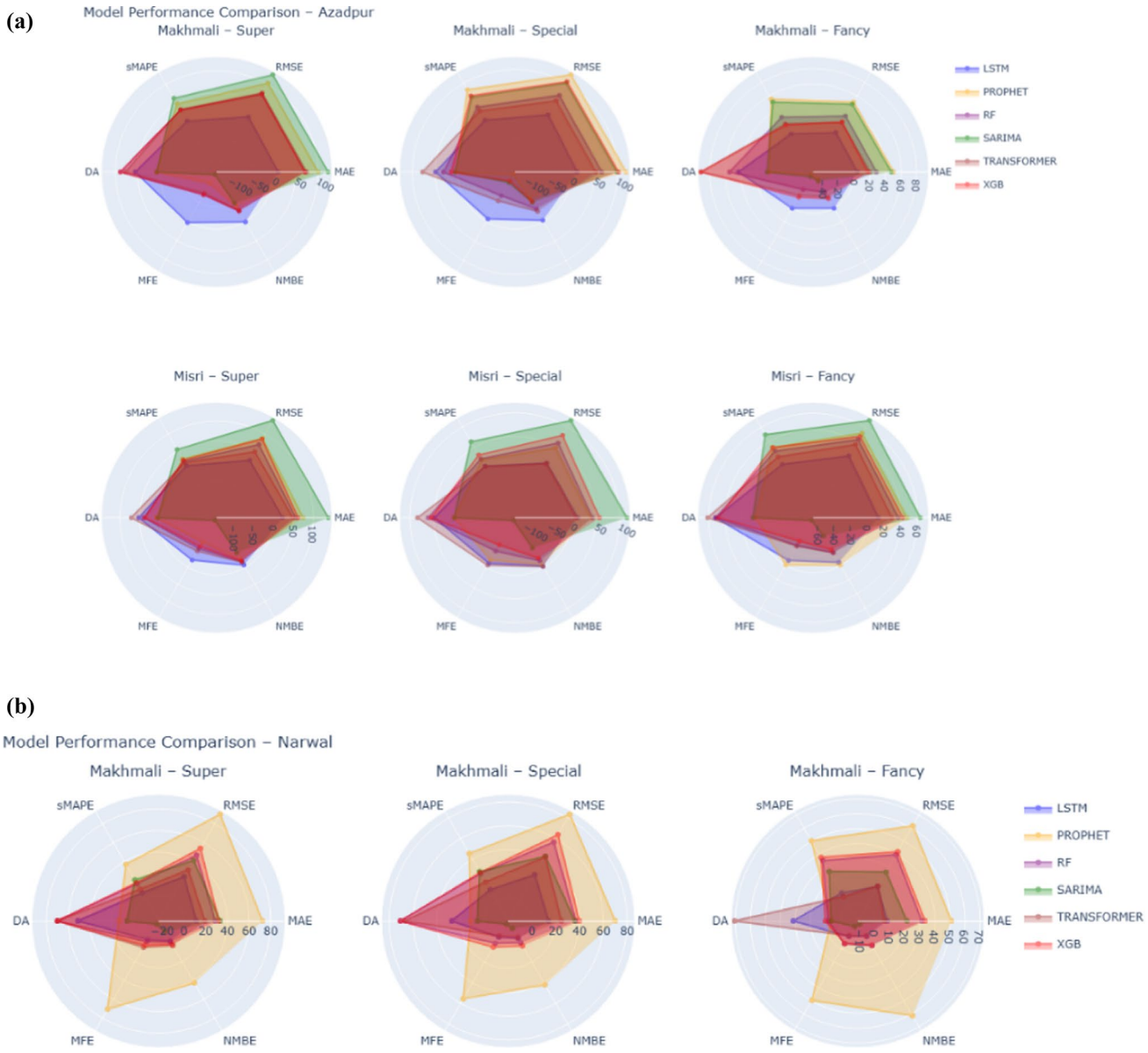


Fig. 14. (**a**) Radar plots showing model performance for the Azadpur market across different variety–grade combinations. (**b**) Radar plots showing model performance for the Narwal market across different variety–grade combinations. (**c**) Radar plots showing model performance for the Ganderbal market across different variety–grade combinations. (**d**) Radar plots showing model performance for the Parimpora market across different variety–grade combinations. (**e**) Radar plots showing model performance for the Shopian market across different variety–grade combinations.

Figure 14a–e show radar plots of how well the five study markets (Azadpur, Ganderbal, Narwal, Parimpore, and Shopian) did at forecasting, broken down by variety–grade combinations. These plots provide the performance of six models, i.e., LSTM, Transformer, RF, XGBoost, SARIMA, and Prophet, at the same time across different evaluation dimensions (MAE, RMSE, sMAPE, DA, MFE, NMBE). There are clear patterns; DL models (LSTM and Transformer) usually make the smallest, most compact polygons, which means they are more accurate and reliable across metrics. On the other hand, traditional statistical models (SARIMA and Prophet) often show wide dispersion and error dominance. Tree-based ensembles (RF, XGB) sometimes do well against each other, but they usually perform worse on datasets with changing grades, like Fancy. Azadpur's Misri–Special and Parimpore's Makhmali–Super showed that Transformer is better in terms of accuracy and DA, while Shopian's Makhmali–Special showed that LSTM is better at balancing performance. This is provided in Appendix Tables 8, 9, 10, 11 and 12, which present the quantitative results for each market and grade, and in Table 4, which highlights the best performers.

DL models, particularly LSTM networks, have repeatedly achieved superior accuracy across diverse forecasting datasets[69,70,72]. Building on this foundation, the attention mechanism in Transformers effectively

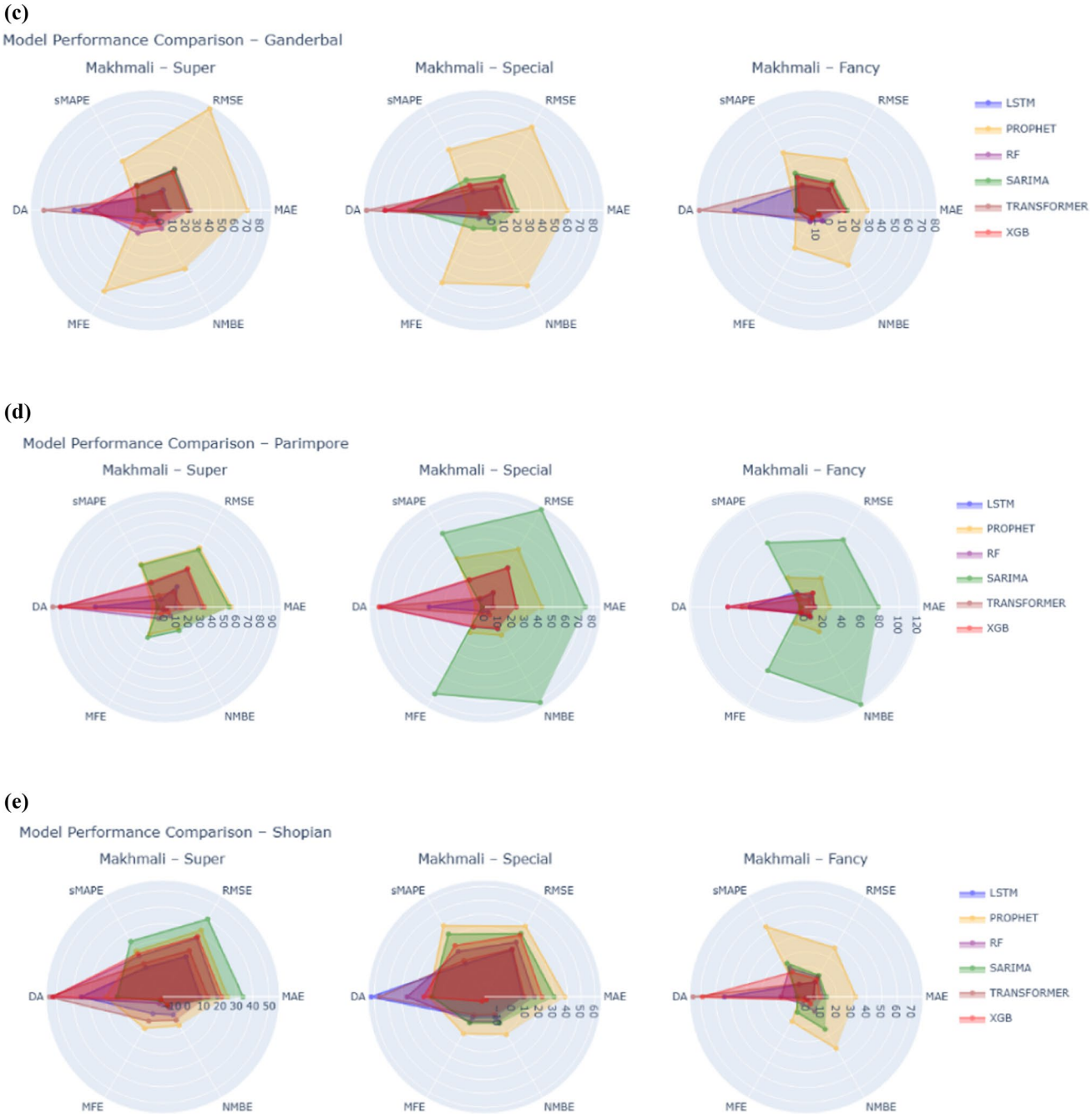


Fig. 14. (continued)

captured complex cherry price patterns even under high variability[25]. In volatile scenarios, some studies have even augmented DL models with regime-switching techniques (e.g., Hidden Markov Models) to bolster forecasting stability[73], further underscoring the importance of resilience in such contexts. By contrast, the traditional statistical models, viz., SARIMA and Prophet, showed higher error margins and lower DA, especially for the higher-grade cherries. This result is in line with prior findings that traditional-based methods struggle with complex or non-stationary series and that Prophet, while helpful in capturing seasonality, can lag in accuracy when abrupt short-term fluctuations dominate[74].

Essentially, without extensive tuning or stationarity assumptions, these older-generation models cannot fully accommodate the intricate patterns in the cherry price data. Similarly, the tree-based ML models (RF and XGBoost) exhibited limitations in modeling the temporal dependencies inherent in price series. Their performance was somewhat inconsistent, reasonable in less volatile conditions, but deteriorating in highly dynamic market regimes. Indeed, studies have noted that XGBoost and similar models often rely on manual feature engineering to incorporate temporal signals and still fall short of recurrent neural networks in chaotic settings[70]. The radar charts strongly support the choice of DL architectures as the most trustworthy tools for

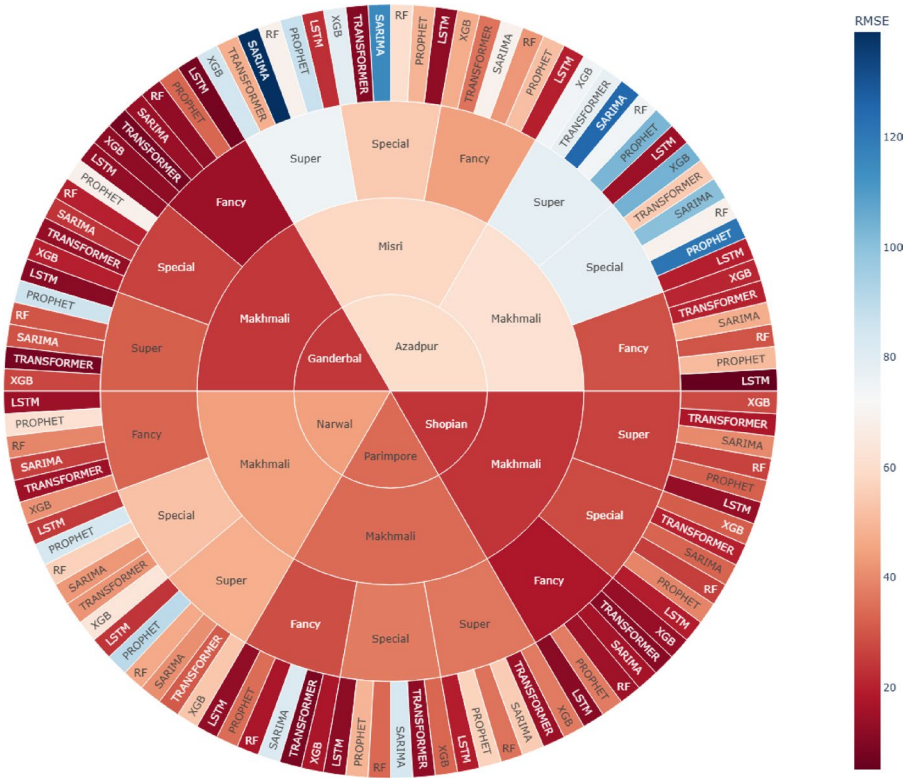


Fig. 15. Sunburst representation of RMSE values across markets and grade combinations for model performance comparison.

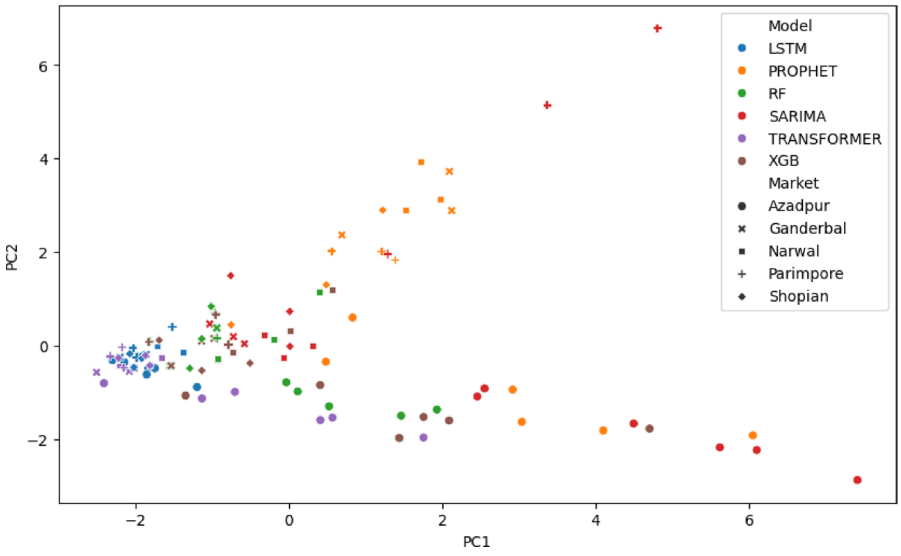


Fig. 16. Principal component analysis (PCA) summarizing variation in model performance metrics across markets and grades.

cherry price forecasting under a range of market conditions. These models not only keep bias low (as seen in near-zero MFE and NMBE) but also maintain high directional reliability, which is crucial for stakeholders. The results here align with the broader consensus in recent forecasting literature, DL models, especially LSTM networks, offer the most reliable and robust performance for commodity price prediction across volatile environments[75].

To summarize performance trends across all markets & grades and ensure they match the radar plots, two small & useful views were added: a Sunburst & a PCA plot. The former (Fig. 15) shows RMSE hierarchically, i.e., Market → Variety → Grade → Model. This makes it easy to see how accurately the forecasts were across the whole

experimental design. The PCA biplot (Fig. 16) shows the six standardized metrics (MAE, RMSE, sMAPE, DA, MFE, NMBE) on two main components, i.e., PC1 & PC2. This gives a joint view of the overall error magnitude (PC1) and the error directionality/bias structure (PC2). Both visuals back up the quantitative model performance metrics with LSTM, making the most compact & consistent cluster. Transformers are competitive but show a little more variability in directional metrics and in classical (SARIMA, Prophet). At the same time, tree-based ensembles (RF, XGB) are spread out across the PCA space, suggesting mixed and less reliable performance profiles. These complementary summaries help confirm that the benefits of DL models are not just due to individual metrics but hold across different types of errors and market-grade situations.

These insights support the broader findings emerging from the time-series forecasting literature where state-of-the-art DL models provide the most reliable and robust performance for commodity market forecasting across a variety of unstable, real-world conditions[68–70,72,74]. The evidence from our results and comparative analysis strongly suggests that stakeholders seeking accurate and dependable price forecasts in cherry markets (and similar agricultural domains) will benefit significantly from leveraging LSTM-based approaches, potentially augmented with hybrid techniques, rather than relying solely on traditional time-series models. Each model has its place, but when it comes to capturing the intricate volatility of seasonal trends, shocks, and non-linear growth in these markets, DL significantly dominates.

Out-of-sample forecasting and real-time validation

For live deployment, the study operationalized the best-performing model (i.e., LSTM) into a lightweight forecasting pipeline tailored to sale-day windows. The deployed architecture used Min–Max scaled targets with a sequence length "L" chosen from [10, 60] via internal validation using MSE. The final model comprised LSTM (100, return_sequences = True) → Dropout(0.2) → LSTM (50) → Dropout(0.2) → Dense (1), optimized with Adam (MSE loss) and early stopping (patience = 5, restore_best_weights = True). Recursive multi-step forecasts were generated for a 15-day horizon by advancing the input window. For dashboard communication, simple heuristic bands (± 1.96 × forecast-path s.d.) were displayed alongside point forecasts. All runs were restricted to achieve sale days, and extensive logging enabled traceability in deployment.

The best-performing model selected during the validation phase was operationalized by deploying a live web-based forecasting system before the 2025 cherry marketing season. Through a secure online interface, market functionaries from designated markets in Delhi and Jammu & Kashmir actively used the system to enter daily price data (see Fig. 17). To generate daily forecasts under realistic, field-level conditions, these real-time entries served as truly unseen, out-of-sample inputs. An end-to-end operational pipeline was completed with forecasts generated automatically and made available via an interactive dashboard. To evaluate the model's generalization performance at the end of the season, predicted prices were compared with actual recorded prices in reverse order. The implemented system provided a scalable model for AI-integrated agri-market intelligence solutions, achieving consistently high predictive accuracy across markets, varieties, and grades. Such practical integration of AI in agriculture is being pursued globally, as researchers advocate the development of free LSTM-driven price forecasting platforms for small producers to support real-time decision-making[76].

The forecasting system remained operationally strong during the cherry marketing season (15 May to 15 July), recording 348 valid entries across 51 active days, for an effective uptime of about 82.3% (Table 6). The system worked well with low latency; the forecasting API responded in about 250 ms, and the supporting endpoints (/preview-data, /latest-prices, /confirm-data) returned results in 5 to 200 ms. Logs showed that core endpoints were 100% reliable and that the dashboard was consistently available, demonstrating that the system was technically stable during real-time deployment. This quick response time is fine for web services;

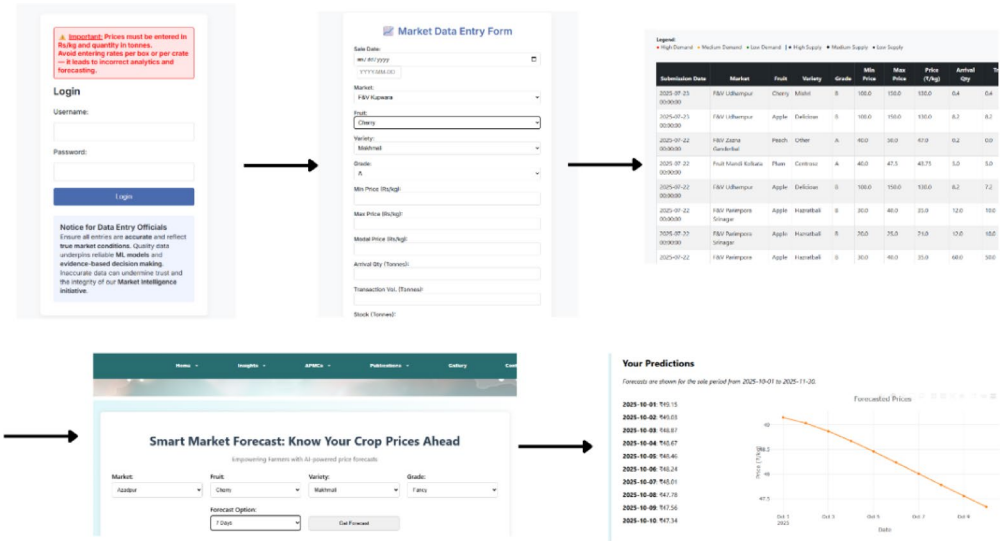


Fig. 17. Overview of the deployed smart forecasting system interface, illustrating the data entry module, market database dashboard, model selection interface, and forecast output.

Dimension	Indicator	Value
Uptime & activity	Active operational days	51 days (15 May–15 July 2025)
	Valid entries recorded	348 entries
	Effective uptime	~82.3%
Latency	Forecast API (/forecast/predict_future) median response	~250 ms
	Other endpoints (/preview-data, /latest-prices, /confirm-data)	5–200 ms
System reliability	HTTP status codes	100% 2xx/3xx for core endpoints; only non-critical 404 s (favicon/robots.txt)
	Dashboard availability	Continuous, no downtime logged
User adoption	Submissions per cherry market	Ranged 1–100 (highest at Shopian = 100, Zazna = 87)
	Submission coverage	Daily inputs from mandi officials; direct farmer usage not yet tracked

Table 6. Operational robustness indicators of the deployed cherry price forecasting system.

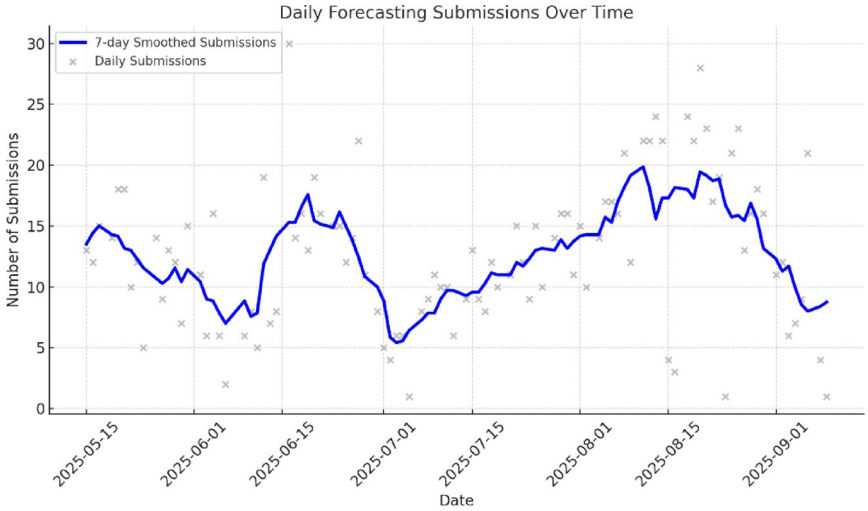


Fig. 18. Time-series of daily forecasting submissions with a 7-day smoothed trend.

if it takes longer than 300 ms, users might stop using the service, but response times of less than a second keep the experience smooth[77]. This level of availability meets or exceeds the usual high-reliability goals, which are often 99.9% uptime for important systems[78]. On the adoption side, officials from selected markets submitted data daily. There were between 1 and 100 entries in each peripheral market, with 87 in Zazna-Ganderbal and 100 in Aglar-Shopian. This ensured that both major production areas and terminal markets were represented, as cherry farming is done chiefly in districts like Ganderbal (which grows about 60% of Kashmir's cherries) and Shopian[79,80]. The platform was technically stable in real time and had sufficient performance headroom to handle regular use without issues, as evidenced by high uptime and quick responses. Figure 18 shows the number of forecasting submissions each day during the harvest period, along with a 7-day smoothed trend line that helps show the system's stability and usefulness. The figure shows that data steadily came in during the cherry marketing season (mid-May to mid-July), and activity then continued for other goods afterward. The continued submission patterns show that institutions adopted the system and that the platform stayed active and reliable after the cherry season.

Even though the pilot didn't track how many farmers used the platform directly, the fact that market officials regularly and proactively submitted data indicates that the system was functional and practical within the current institutional framework. Market officers made the forecasting tool a regular part of their workday, a strong sign that the system is valuable and easy to use in a real business setting. This kind of institutional adoption is essential for farmers to benefit in the long run. Studies have shown that accurate price forecasts help farmers reduce market risks and help officials run agricultural markets better[81].

Table 7 summarizes the performance of the deployed LSTM model and shows clear stratification across market-variety-grade combinations, with consistently low error metrics and high directional accuracy. The model achieved 95.8% accuracy, a remarkably low sMAPE of 4.31%, and minimized MAE and RMSE of 6.28 and 9.45, respectively, in the Azadpur market for Makhmali Super cherries. Exceptionally, LSTM applications in other crop markets reported percentage errors in the high single digits to low teens[5], underscoring the exceptional precision of the 4.31% sMAPE obtained in the present study. Similarly, the forecasts in the Narwal Fancy and Shopian Special grades demonstrated strong agreement with observed trends, registering low bias (NMBE under 4%) and high model accuracy (92.8% and 93.95%), confirming the robustness and dependability of the deployed

Market	Variety	Grade	MAE	RMSE	sMAPE	DA (%)	MFE	NMBE (%)	Accuracy (%)
Azadpur	Makhmali	Super	6.28	9.45	4.31	50	4.07	2.72	95.8
		Special	8.03	9.51	7.56	50	4.08	3.85	92.42
		Fancy	10.08	12.17	12.2	57.14	10.08	12.61	87.4
	Misri	Super	30.89	34.04	11.99	95.45	30.89	12.33	87.67
		Special	12.02	12.89	5.5	4.55	12.02	5.63	94.37
		Fancy	10.3	11.54	8.4	50	10.3	8.64	91.36
Narwal	Makhmali	Super	7.49	8.44	5.38	31.82	3.96	2.87	94.58
		Special	7.76	9.19	6.86	31.82	-4.09	-3.58	93.22
		Fancy	4.84	6.21	7.1	22.73	4.84	7.2	92.8
Ganderbal		Super	11.49	12.56	8.72	16.67	10.26	8.01	91.03
		Special	22.84	23.27	25.3	33.33	14.6	17.1	73.25
		Fancy	5.43	6.32	7.83	33.33	-4.1	-5.83	92.28
Parimpora		Super	8.88	9.89	6.84	35.71	0.33	0.26	93.15
		Special	5.25	5.63	5.25	21.43	2.47	2.49	94.7
		Fancy	8.11	14.13	9.9	14.29	-8.1	-10.38	89.6
Shopian		Super	7.51	9.31	6.09	27.27	-0.71	-0.58	93.88
		Special	5.19	6.11	6.02	13.64	3.03	3.53	93.95
		Fancy	16.37	17.07	25	22.73	16.37	28.41	71.58

Table 7. Out-of-sample forecasting accuracy by market and crop segment (2025 test data).

model in the pragmatic settings. This level of robustness aligns with global findings that DL methods can handle nonlinear market fluctuations and maintain high accuracy[82], confirming the model's reliability.

The Shopian Fancy case is an intriguing exception. However, its MFE and NMBE show a propensity to overestimate, with values of 16.37 and 28.41, respectively. Its RMSE and MAE are high (17.07 and 16.37, respectively), indicating volatility in the raw error values. Other studies have addressed similar issues by augmenting LSTM models with statistical or decomposition techniques to capture abrupt swings, for instance, combining LSTM with ARIMA for linear trends[83,84] or applying seasonal-trend decomposition (STL) before LSTM to handle seasonal spikes[5]. These approaches could potentially mitigate the overshooting observed in Shopian Fancy. Despite this, the grade maintained an accuracy of 71.58%, suggesting that the forecasting consistency in that sub-market was acceptable but not very reliable. Similarly, performance in Ganderbal varied greatly by grade. The Special grade encountered significant difficulties, with the highest sMAPE at 25.3%, suggesting potential structural deviations or increased volatility in the price signal during the test period. In contrast, the Fancy grade saw a well-balanced forecast with MAE at 5.43 and sMAPE at 7.83%. In addition to volatility, however, the relatively weaker performance in Shopian Fancy and Ganderbal Special is plausibly linked to structural deviations in the underlying market signals, such as late-season regime shifts, thinner trading volumes, and irregular reporting patterns. These conditions likely led to data sparsity and abrupt changes that a single-regime LSTM, trained on comparatively short sequences, could not fully capture. By contrast, the Narwal and Azadpur models consistently maintained high accuracy (above 92% across multiple segments), indicating strong generalization capacity in line with market-level trends. Although the Special grade continued to function reliably, the models at Parimpora showed slightly more dispersion, with the Fancy grade accuracy falling to 89.6% due to a higher negative bias (NMBE of −10.38). In addition to confirming the chosen models' resilience, these test results show how well they can adapt to real-time changes in crop prices across a variety of geographic areas.

The model maintained a smoother output, indicating conservative estimation in noisy conditions, a known behavior of regularized LSTM architectures trained with dropout and early stopping, even though some markets, such as Ganderbal, showed higher variance in the actual prices. The model's ability to maintain fine-grained pricing differentiation, particularly between Super, Special, and Fancy classifications, is further supported by a comparative analysis of predictive accuracy across grades. The Bland–Altman plots (Fig. 19) provide a reliable diagnostic for agreement between the test and validation phases. With mean differences centered close to zero and the residuals primarily clustering within the 95 percent limits of agreement, there is no evidence of systemic bias during the shift from training-validation to unseen test deployment.

The bivariate scatter plots (Fig. 20) compare test and validation performance across RMSE, sMAPE, MFE, and NMBE and show a strong linear relationship, with narrow confidence ellipses, indicating minimal performance degradation and high model stability. These findings together demonstrate that the proposed LSTM-based architecture is a strong contender for implementation in real-time agricultural market advisory systems, as it is both statistically consistent and operationally robust. Consistent with earlier reports highlighting the superior performance of deep learning models in complex forecasting tasks[85,86], it is positioned as a strong decision-support tool within the larger agri-market intelligence framework due to its ability to capture both inter-grade variations and market-specific volatility under real-world conditions.

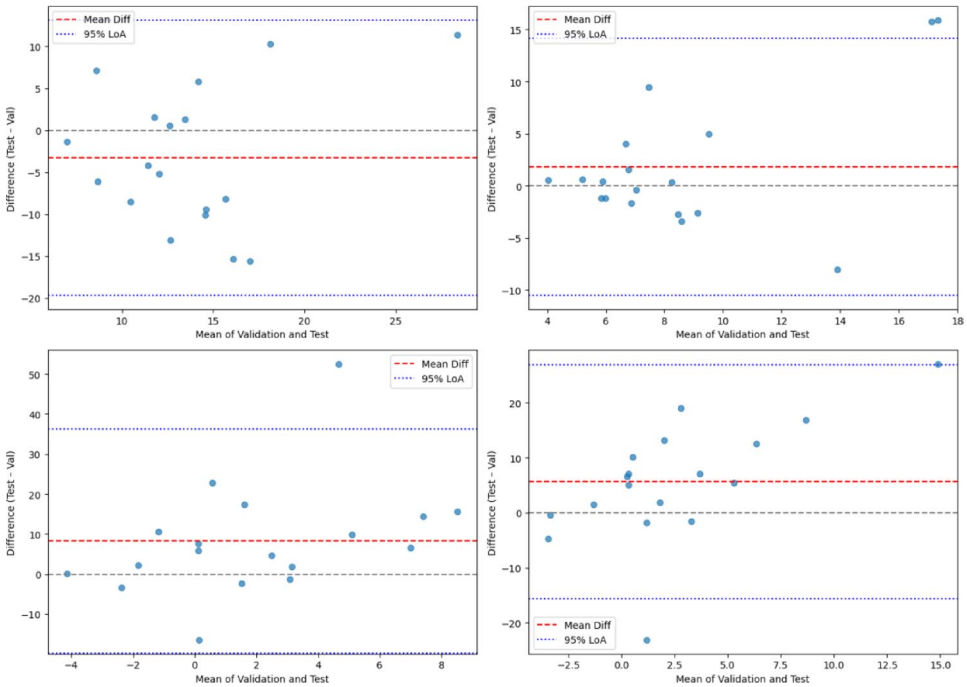


Fig. 19. Bland–Altman analysis comparing the agreement between forecast errors from the validation and test datasets.

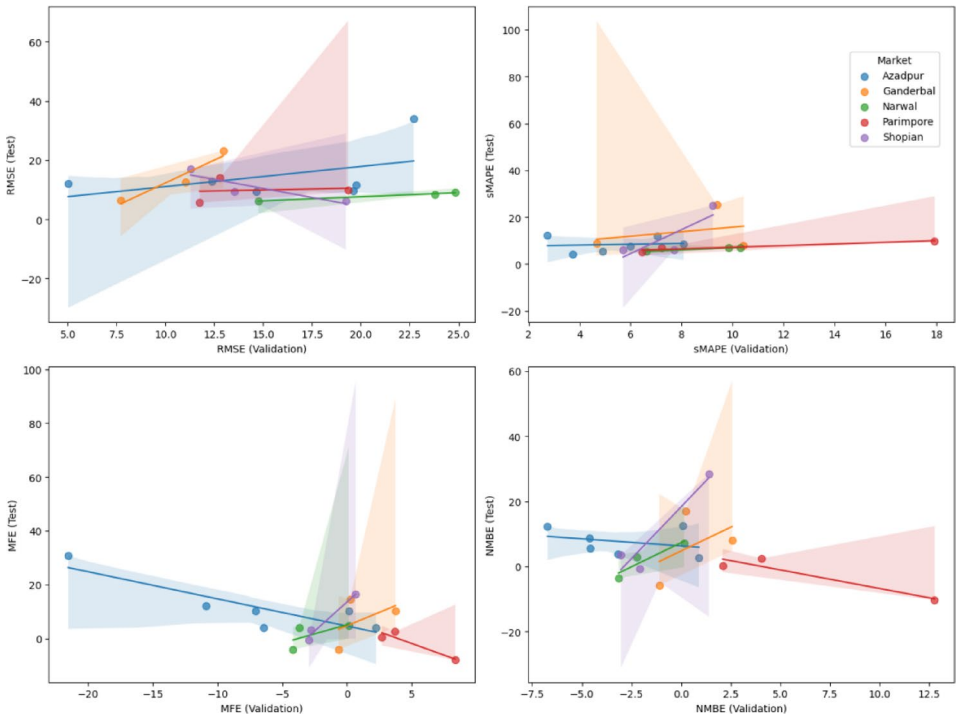


Fig. 20. Comparison of validation- and test-set performance metrics across markets to assess metric concordance.

Policy implications and limitations

The results of this research have significant policy implications for the formulation of agricultural market intelligence frameworks in India, particularly for high-value perishable commodities such as cherries. DL models, especially LSTMs, have been shown to capture nonlinear price dynamics better and were successfully used in a real-time setting during the 2025 season. The study results showed that it is possible to incorporate

advanced forecasting tools into current national and state-level horticultural trade initiatives in the hub and terminal markets. The proposed system aligns with Jammu and Kashmir's Digital Agriculture Mission, the Electronic National Agricultural Market (e-NAM), and the Holistic Agriculture Development Programme (HADP). Adding real-time AI forecasting to these programs could make price intelligence a digital public good, thereby opening price discovery, reducing price volatility, and helping the Government of India achieve its long-term goal of doubling farmers' incomes. The study provides data-driven evidence that precise price forecasting enables farmers, traders, and policymakers to make proactive, informed decisions, increasing transparency, reducing post-harvest losses, and improving market timing and inventory management. Such model architectures can identify intricate nonlinearities and provide actionable intelligence on real-time market dynamics, potentially significantly reducing price risk and supporting income stability. The results support the developed forecasting system through its usefulness and by identifying DL decision support systems as a game-changing tool for boosting market efficiency, strengthening the resilience of food systems, and promoting inclusive agricultural development, especially in data-constrained economies.

One of the most significant problems in horticultural marketing is the risk posed by price volatility, which often leaves smallholders unaware of the value of a season's harvest. The system tested in the study provided stakeholders with insights to mitigate this risk and help them make effective interventions based on short-term & accurate forecasts. Also, forecasting can inform the timing and scale of the Market Intervention Scheme (MIS), when to release funds from the Price Stabilisation Fund (PSF), and when to provide targeted subsidies for cold storage and transport in case of a harvest glut. Through this novel initiative, the state can shift from reactive crisis management to proactive crisis management, stabilizing incomes and lessening market shocks. This system can also be essential for improving market governance and openness. Information asymmetry remains a significant factor in Indian agricultural markets. Farmers usually accept the price, while traders and middlemen have substantial power to negotiate. Putting forecasting outputs into APMCs, horticulture directorates, and NABARD- or FPO-led platforms can make daily price information available to everyone, helping farmers make smart decisions about when and where to sell their produce. The framework's ability to grow provides policy designers with more options during any crisis.

Moreover, the architecture can easily be applied to other high-value perishable goods, such as apples, plums, pears, and vegetables, which are subject to volatility, perishability, and broken marketing channels. By connecting forecast dashboards to state market networks, farmer advisory apps like Kisan Sarathi, and new digital infrastructures like AgriStack, AI-enabled intelligence can help farmers and traders make sound decisions daily. Also, this system can be replicated and scaled up in other states across different agricultural commodities, thus creating a national system of innovative data-driven agricultural marketing structure. These systems can also be an essential source of information for policy analysts seeking to address climatic variations that accompany market volatility. This integration would make supply chains more resilient in the face of real-world challenges.

However, these advantages can be leveraged only when institutional mechanisms adopt AI-based forecasting from the grassroots level to the state level, and ultimately to export stages. Public–Private Partnerships (PPPs) may be encouraged to use such developed platforms to improve their purchasing and shipping plans. This kind of institutional convergence would ensure that AI-powered forecasts aren't just research prototypes but also have industrial and commercial applications. This system can also be advantageous for integrating smallholders, tribal growers, and women farmers in remote areas into extended value chains and for providing them with timely information about forecasts through easy-to-use channels such as SMS alerts, IVR systems, or community-based digital kiosks. Through these pragmatic uses of the developed models, the multiple stakeholders, especially policymakers, can promote inclusive growth and development across the various landholdings, regions, and markets. Besides, it will also help bridge the digital divide and social gaps in India, especially in the areas with resource constraints. The developed system provides a robust Data Driven Intelligence System (DDIS) to mitigate price volatility, stabilize farmer incomes, enhance market transparency, foster resilience to external shocks, and promote inclusive development.

Even with these strengths, some weaknesses limit the models' generalizability & scalability. The current implementation used only endogenous price histories and did not include any external factors, such as weather shocks, arrival data, transport bottlenecks, or policy changes. This omission limits the model's capacity to predict structural breaks or sudden shocks that frequently induce volatility in fresh produce markets. The system was successfully tested with officials in selected markets and needs to be tested at the farm-gate or grassroots level. DL models may not be widely adopted in low-resource or rural contexts due to their computational complexity and resource intensity. Interpretability and trust among grassroots stakeholders are also hampered by the risk of overfitting and by a lack of transparency into model behavior. The models may not be as applicable to other commodities or regions without retraining, as the training dataset was geographically and temporally confined to the J&K and Delhi markets, despite its high intra-seasonal resolution. Finally, to increase robustness across variables, hybrid ensemble approaches and more detailed, cross-sectoral data integration are required, as evidenced by inconsistent performance across some grades.

Conclusion

This study illustrated the novel real-time implementation of DL models for forecasting in horticultural commodities in India, thereby bridging a significant gap in the literature concerning high-value & perishable crops. The research systematically benchmarks six models, i.e., SARIMA, Prophet, Random Forest, XGBoost, LSTM, and Transformer, across five major markets, demonstrating that deep learning approaches, especially LSTM and Transformer, consistently surpass traditional statistical and tree-based methods. These models maintained their market-variety-grade-wise fidelity and high predictive accuracy, even when markets were unstable and competing commodities were present. The distinctive contribution lies in implementing the LSTM model within a live online system during the 2025 cherry season, enabling real-time out-of-sample forecasting

Variety	Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Makhmali	Super	LSTM	9.50	14.64	3.74	54.29	2.24	0.89
		PROPHET	100.04	102.89	48.89	5.71	-100.04	-39.58
		RF	70.61	74.35	32.73	88.57	-70.61	-27.93
		SARIMA	122.79	124.87	63.66	5.71	-122.79	-48.57
		TRANSFORMER	74.29	77.39	33.98	77.14	-74.29	-29.39
		XGB	70.19	74.67	32.94	88.57	-70.19	-27.77
	Special	LSTM	11.41	19.61	6.00	48.57	-6.44	-3.19
		PROPHET	118.67	121.06	83.16	2.86	-118.67	-58.81
		RF	65.32	69.11	39.14	31.43	-65.32	-32.37
		SARIMA	98.63	100.20	64.19	5.71	-98.63	-48.87
		TRANSFORMER	52.35	55.00	29.36	77.14	-52.35	-25.94
		XGB	102.15	103.94	68.09	14.29	-102.15	-50.62
	Fancy	LSTM	3.21	5.02	2.74	42.86	0.12	0.10
		PROPHET	50.13	50.74	55.18	2.86	-50.13	-43.28
		RF	27.92	29.76	28.06	54.29	-27.92	-24.11
		SARIMA	47.07	47.48	50.87	5.71	-47.07	-40.64
		TRANSFORMER	19.43	20.06	18.19	91.43	-19.43	-16.77
		XGB	16.38	21.08	16.36	91.43	-16.38	-14.14
Misri	Super	LSTM	21.54	22.70	7.05	51.61	-21.54	-6.73
		PROPHET	73.62	87.31	27.04	6.45	-73.62	-23.00
		RF	55.28	68.84	20.22	38.71	-55.08	-17.21
		SARIMA	137.29	139.18	54.54	3.23	-137.29	-42.88
		TRANSFORMER	47.70	47.92	16.09	70.97	-47.70	-14.90
		XGB	61.09	84.80	23.88	35.48	-61.09	-19.08
	Special	LSTM	11.38	12.36	4.90	48.39	-10.87	-4.57
		PROPHET	38.44	48.90	18.83	6.45	-18.82	-7.92
		RF	42.40	60.56	22.00	61.29	-40.20	-16.91
		SARIMA	114.96	115.42	63.70	6.45	-114.96	-48.37
		TRANSFORMER	8.04	10.40	3.34	83.87	-5.89	-2.48
		XGB	57.14	79.29	31.62	54.84	-55.36	-23.29
	Fancy	LSTM	12.29	19.77	8.08	51.61	-7.07	-4.61
		PROPHET	47.04	51.82	31.82	3.23	-0.87	-0.57
		RF	36.54	42.62	26.51	51.61	-28.04	-18.27
		SARIMA	63.08	69.42	49.63	6.45	-63.08	-41.11
		TRANSFORMER	28.38	35.82	18.52	61.29	-27.08	-17.65
		XGB	41.86	47.15	31.58	48.39	-33.96	-22.13

Table 8. Detailed model-wise forecasting performance for the Azadpur market across cherry grades.

based on daily market submissions. Comparative validation confirmed with temporal consistency, low bias drift, and robust adaptation to intra-seasonal volatility, highlighting the model's practical reliability. The analysis also showed how grade differentiation and market structure affect price changes, supporting the economic rationale for market-variety, grade-based stratification. The system provides stakeholders, such as farmers, traders, and policymakers, with a novel, tested tool for making real-time decisions. The study developed a credible framework for integrating AI-based forecasting into India's agricultural marketing system by assessing its robustness through several statistical tests and visual diagnostics. This method is a scalable plan for bringing AI-enabled market intelligence to other high-value crops in the future. This will help keep income stable, markets more transparent, and markets more resilient, especially for perishable goods.

Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Super	LSTM	6.73	11.02	4.66	53.66	3.74	2.57
	PROPHET	69.23	86.04	37.54	2.44	67.14	46.15
	RF	23.39	29.93	15.22	36.59	12.99	8.93
	SARIMA	21.77	29.40	14.61	2.44	-5.40	-3.71
	TRANSFORMER	5.58	8.00	3.83	78.05	0.09	0.06
	XGB	21.18	26.92	14.07	46.34	7.07	4.86
Special	LSTM	8.64	12.96	9.38	48.78	0.23	0.24
	PROPHET	58.88	68.67	47.92	4.88	58.88	61.59
	RF	14.04	19.95	15.21	70.73	-4.18	-4.37
	SARIMA	19.02	23.65	20.55	51.22	9.27	9.69
	TRANSFORMER	10.91	13.20	11.95	85.37	-1.28	-1.33
	XGB	14.36	19.67	15.56	70.73	-4.33	-4.53
Fancy	LSTM	6.18	7.71	10.43	51.22	-0.65	-1.10
	PROPHET	27.92	33.17	39.77	4.88	22.17	37.46
	RF	11.15	12.46	19.15	4.88	-0.80	-1.35
	SARIMA	12.70	14.14	21.80	4.88	-4.72	-7.98
	TRANSFORMER	7.07	7.82	11.92	78.05	-3.32	-5.61
	XGB	10.33	12.26	17.69	2.44	-4.00	-6.76

Table 9. Detailed model-wise forecasting performance for the Ganderbal market across cherry grades.

Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Super	LSTM	11.43	23.78	6.61	50.94	-3.71	-2.23
	PROPHET	72.39	90.05	36.89	13.21	70.55	42.45
	RF	28.75	46.59	16.19	69.81	-1.44	-0.87
	SARIMA	33.43	40.96	20.35	5.66	-20.07	-12.07
	TRANSFORMER	19.32	30.90	10.30	69.81	4.67	2.81
	XGB	32.19	53.67	17.33	69.81	2.57	1.55
Special	LSTM	13.93	24.83	10.29	27.45	-4.17	-3.15
	PROPHET	69.98	83.46	45.51	11.76	55.48	41.90
	RF	36.79	56.56	26.12	70.59	1.63	1.23
	SARIMA	35.92	42.78	28.04	5.88	-16.96	-12.81
	TRANSFORMER	24.78	42.16	17.85	66.67	-5.31	-4.01
	XGB	39.77	63.68	27.02	70.59	5.42	4.10
Fancy	LSTM	7.87	14.75	9.85	31.37	0.13	0.16
	PROPHET	50.86	61.77	50.24	5.88	49.51	61.44
	RF	30.99	39.36	34.87	7.84	5.81	7.21
	SARIMA	21.05	25.86	26.40	5.88	-7.62	-9.45
	TRANSFORMER	6.11	15.10	7.03	70.59	-0.36	-0.45
	XGB	33.21	41.60	37.38	9.80	5.58	6.93

Table 10. Detailed model-wise forecasting performance for the Narwal market across cherry grades.

Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Super	LSTM	8.48	19.34	7.22	58.33	2.70	2.09
	PROPHET	55.00	56.83	41.25	10.42	26.27	20.32
	RF	30.15	35.64	22.79	87.50	11.14	8.62
	SARIMA	53.00	54.94	40.30	6.25	29.55	22.86
	TRANSFORMER	12.71	14.33	10.68	93.75	1.98	1.54
	XGB	32.31	37.12	23.93	87.50	6.60	5.10
Special	LSTM	5.67	11.75	6.45	41.30	3.69	4.05
	PROPHET	42.00	49.31	41.36	6.52	22.03	24.19
	RF	23.13	33.67	22.66	76.09	17.36	19.06
	SARIMA	74.32	83.27	62.79	2.17	74.32	81.61
	TRANSFORMER	6.28	12.37	7.44	78.26	6.15	6.76
	XGB	23.29	33.24	23.34	78.26	16.56	18.19
Fancy	LSTM	11.12	12.80	17.92	57.45	8.36	12.77
	PROPHET	26.32	34.48	35.47	4.26	19.77	30.22
	RF	10.22	17.08	14.93	80.85	7.55	11.54
	SARIMA	77.57	81.58	78.05	4.26	77.57	118.58
	TRANSFORMER	5.61	8.68	9.36	65.96	5.61	8.58
	XGB	10.03	17.04	14.61	80.85	7.45	11.39

Table 11. Detailed model-wise forecasting performance for the Parimpora market across cherry grades.

Grade	Model	MAE	RMSE	sMAPE	DA	MFE	NMBE
Super	LSTM	7.91	13.54	5.71	34.78	-2.94	-2.07
	PROPHET	24.77	31.99	17.53	19.57	7.40	5.21
	RF	18.61	26.52	13.88	52.17	-10.32	-7.27
	SARIMA	34.18	39.96	24.18	13.04	-12.26	-8.64
	TRANSFORMER	12.66	17.68	8.60	54.35	2.25	1.59
	XGB	21.26	27.79	15.48	52.17	-12.86	-9.06
Special	LSTM	7.54	19.23	7.70	62.00	-2.80	-3.03
	PROPHET	38.24	39.02	39.47	20.00	10.66	11.52
	RF	16.08	25.67	17.94	36.00	0.70	0.76
	SARIMA	30.38	32.85	32.37	18.00	1.88	2.03
	TRANSFORMER	8.92	19.98	10.23	56.00	-4.53	-4.90
	XGB	21.61	31.49	22.86	24.00	-16.27	-17.58
Fancy	LSTM	4.74	11.28	9.23	54.17	0.67	1.39
	PROPHET	32.64	37.36	53.62	4.17	19.01	39.47
	RF	11.27	15.10	22.10	16.67	5.32	11.05
	SARIMA	13.24	16.28	25.57	6.25	11.95	24.81
	TRANSFORMER	4.82	11.19	9.88	75.00	1.06	2.20
	XGB	9.21	13.60	18.77	68.75	2.57	5.33

Table 12. Detailed model-wise forecasting performance for the Shopian market across cherry grades.

Data availability

An anonymised, cleaned sample dataset (Shopian market, J&K), along with minimal reproducibility code covering the data pipeline, training/testing, and forecasting scripts, and requirements.txt, has been deposited on Zenodo: Gul, A. (2025). *Sample Dataset and Minimal Reproducibility Package for Cherry Price Forecasting (Shopian Market, J&K)* (Version A1). Zenodo. [https://doi.org/10.5281/zenodo.17075935](https:/doi.org/10.5281/zenodo.17075935) . The full raw market registers cannot be shared publicly due to institutional restrictions but can be requested from the corresponding author in line with data-sharing policies.

Appendix

See Tables 8, 9, 10, 11 and 12.

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F. A. Shaheen: Conceptualization, Investigation, Administering of project and overall control, Writing – Review & Editing. Aqib Gul: Methodology, Data Curation, Formal Analysis, Visualization, Writing – Original Draft . Nazir Ganai : Conceptualization, Investigation, Review & Editing. Masroor Majid: Writing – Review & Editing, Mudasir Rashid: Data collection, scanning & digitization, Abid Sultan: Result verification,Policy advisory.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
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